

Chapter 9: Search Engines for OSINT

Coverage, Ranking, and Why One Engine Is Never Enough

Companion Workbook to the Instructional Video

How to Use This Workbook

This is the written companion to the instructional video on search engines. It works as a follow-along guide, a standalone textbook chapter, and a practice manual.

Where this chapter sits in the series. Chapter 3 covered the nineteen source categories and made a point that this chapter develops: a search engine is a tool that indexes part of one category, not a source in itself. Chapter 4 covered the collection process, Chapter 6 the documentation discipline, and Chapter 8 the operational security that applies to your own searching. **This chapter is about using search engines competently**, which turns out to be a much larger subject than most people expect.

A warning about volatility. This is the most perishable chapter in the series. Search engines change their operators, their partnerships, their index sources, and their existence. Engines that were significant when this chapter was written may be gone or transformed by the time you read it, and partnerships between engines change without announcement.

The chapter is therefore written to teach **method rather than tool**. Every engine section explains what kind of thing the engine is and what it is good for, and treats specific syntax as illustrative. Where a section describes a partnership or an index arrangement, treat it as a snapshot to verify rather than a permanent fact. Section 1.1.6 makes this explicit and gives you the way to check it yourself.

Time estimate:

Component	Estimated time
Video	60 to 75 minutes
Reading material	5 to 6 hours
Knowledge assessment	1 hour
Critical thinking exercises	2 hours
Labs 1 through 5	12 to 17 hours
Capstone assignment	6 to 9 hours
Total	27 to 36 hours

Materials: a computer with internet access and a browser. Labs 1 and 3 involve live searching; Labs 2 and 4 use provided result packets so that everyone is graded against the same material. Every lab has a path that works if a particular engine is unreachable from where you are.

The Rules Still Apply

Rule 1: No private individuals. Rule 2: Publicly available means genuinely public. Rule 3: Document as you go. Rule 4: No dark web access. Rule 5: Automation is constrained. Rule 6: A relationship is a claim, and every claim needs a basis. Rule 7: If it is not written down, it did not happen. Rule 8: Lawful is the floor, not the ceiling. Rule 9: OPSEC never buys access.

This chapter adds the rule that governs everything in it:

Rule 10: Absence of results is a statement about the index, not about the world.

A search that returns nothing tells you that this engine's crawler has not indexed a page containing those terms in a form its ranking system chose to return to you, from your location, at this moment. It does not tell you the thing does not exist. Chapter 3 taught you to attach a coverage boundary to every negative finding; this chapter explains, mechanically, why that boundary is so much narrower than investigators assume.

Rule 5 also has a specific application here. Automated querying of search engines breaches the terms of every major engine, is detected routinely, and is prohibited in this course. Every lab is performed manually.

Learning Objectives

Knowledge objectives (you can explain it):

1. Explain how a search engine crawls, indexes, ranks, and retrieves, and what can go wrong at each stage.
2. Explain why two engines return different results for the same query, naming at least five causes.
3. Distinguish an engine with an independent index from one that aggregates or licenses another's results.
4. Distinguish a search engine from a metasearch engine and from a privacy proxy.
5. Explain personalization, localization, and freshness bias, and their consequences for investigative work.
6. Describe the characteristics, strengths, and limitations of each major engine covered.
7. Explain why regional engines matter and what they hold that global engines do not.
8. Explain the privacy differences between engines and what each does and does not protect.
9. Explain what search operators do conceptually, and why specific syntax must always be verified.
10. Explain what a search engine cannot reach, connecting to Chapter 3's deep web material.

Application objectives (you can do it):

1. Construct a search plan for an investigative question, selecting engines with stated reasons.
2. Execute the same query across multiple engines and document differences in ranking, coverage, freshness, and language.
3. Identify unique findings attributable to a specific engine.
4. Document searches to the Chapter 6 standard, including negative results with coverage boundaries.
5. Use regional engines to reach material global engines miss.
6. Build a decision matrix that selects engines by objective, language, geography, privacy need, and expected source type.

Analysis objectives (you can judge it):

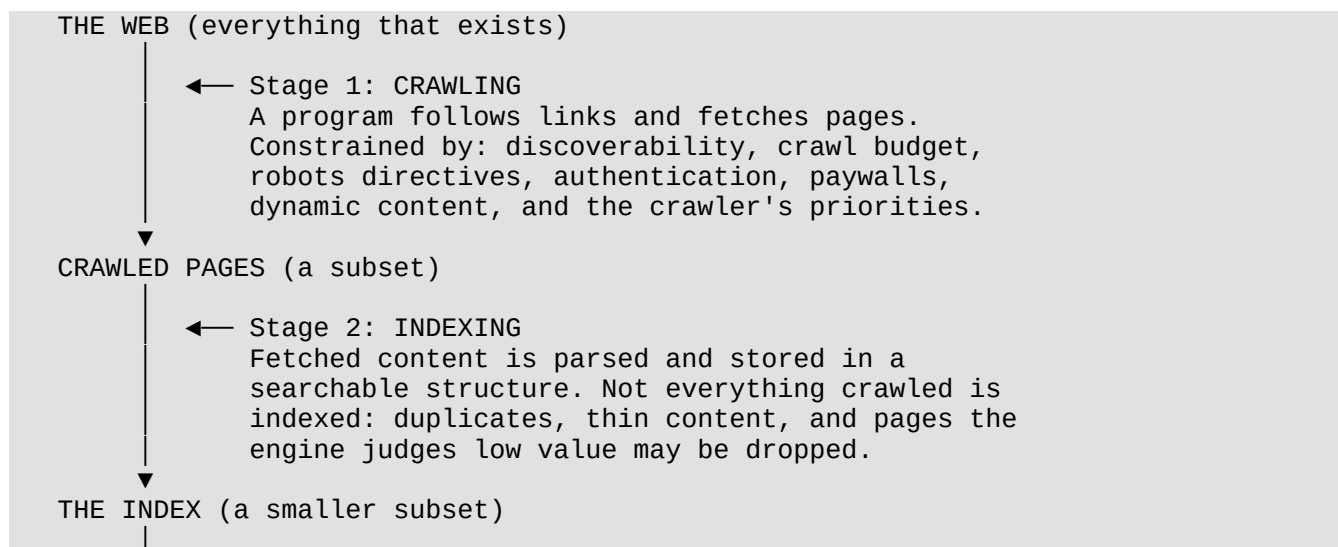
1. Assess whether a negative result reflects absence or index coverage.
 2. Evaluate whether apparently corroborating results are independent or derive from one index.
 3. Judge which engine's result is more reliable for a given claim, and why.
 4. Recognize when a search engine is the wrong tool and a structured source is required.
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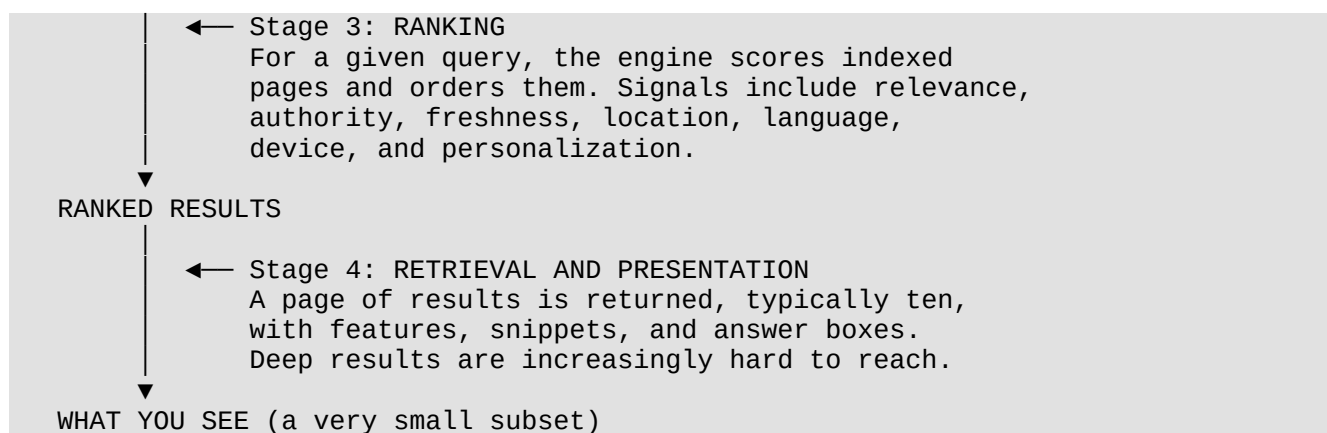
Part 1: Reading Material

1.1 Introduction to Search Engines

1.1.1 The Four Stages

Everything an investigator needs to understand about search engine behaviour follows from four stages. Each stage discards information, and the discarding is where your findings are won and lost.





The investigative consequence: what you see is four filters away from what exists. Any of the four can be the reason your search failed, and each has a different remedy.

Stage	Why your search failed	Remedy
Crawling	The page was never fetched: not linked, blocked, behind a login, or in a format the crawler skipped	Go to the source system directly (Chapter 3); try an engine with a different crawler
Indexing	Fetched but not kept	Try another engine; try an archive
Ranking	Indexed but buried below where you looked	Better query; operators; different engine with different ranking
Retrieval	Ranked but not displayed to you, because of your location, language, or personalization	Change locale settings; use a different engine; use a session without personalization

1.1.2 Why Two Engines Differ

Five causes, and students should be able to name all five.

1. Different indexes. The engines crawled different pages. An independently indexed engine may hold pages another never fetched, and vice versa.

2. Different ranking. Even with identical indexes, different scoring produces different orders, and order determines what you actually see.

3. Different localization. Your apparent location changes both the results and the ranking. The same query from two countries returns materially different pages.

4. Different personalization and session state. History, signed-in status, and prior behaviour shift results on some engines.

5. Different freshness policies. Engines recrawl at different rates and weight recency differently, so a page published this morning may exist in one index and not another.

A sixth cause that surprises people: the same engine returns different results to the same person at different times, because indexes and ranking models change continuously. **A search is not reproducible.** This is why Chapter 6's search log records the engine, the exact query, the filters, the date and time, and the result count, and why a screenshot of the results page is itself evidence.

1.1.3 What Search Engines Cannot Reach

Chapter 3 established that the deep web is where most authoritative material lives. Restated mechanically:

Not reachable by a general crawler	Examples
Content behind a query form	Court case systems, corporate registries, most databases
Content behind authentication	Subscription archives, member areas, webmail
Content excluded by directives	Pages a site asks crawlers not to fetch
Content the crawler cannot render or parse	Some dynamic interfaces, some document formats
Content nobody links to	Orphan pages, unlisted documents
Content published and then removed	Only archives hold it (Chapter 3)
Content in a format or language the engine deprioritizes	Regional and minority-language material

The consequence for Rule 10: an empty search result is consistent with all seven of these, and with the thing genuinely not existing. Only one of those eight possibilities is "it does not exist," and the investigator's job is to say which possibilities their search actually excluded.

1.1.4 Independent Indexes and Aggregated Results

The most consequential structural fact about the search landscape, and the one that determines whether two agreeing results are corroboration or an echo.

Type	Description	Investigative meaning
Independent index	The operator runs its own crawler and builds its own index	Genuinely different coverage; a real second opinion
Licensed or partnered results	The interface is its own; the results come substantially from another operator's index	Different presentation, largely the same coverage
Hybrid	Own crawler for part of the web, supplemented by a partner for the rest	Partial independence; the proportion matters and is rarely disclosed precisely
Metasearch	Queries several engines and merges the results	Breadth of presentation, no index of its own
Privacy proxy	Passes your query to another	The other engine's coverage,

Type	Description	Investigative meaning
	engine without your identifiers	with better privacy

The corroboration trap. If three engines return the same result and all three draw on one index, you have one source presented three times. This is Chapter 2's circular reporting in a new form, and it is invisible unless you know which engines are independent.

How to find out for yourself, since the arrangements change:

- Engines that maintain their own crawler usually say so publicly and describe their crawler's user agent.
- Compare results for an obscure query: engines sharing an index return strikingly similar result sets, including similar snippets and similar ordering.
- Look for the presence or absence of pages you know exist in one index and not another.
- Check the engine's own documentation and any published partnership announcements.

1.1.5 Personalization, Localization, and Freshness

Factor	What it does	Investigative problem	Mitigation
Personalization	Adjusts results based on your history and account	You see a version of the web shaped by your prior behaviour, and cannot tell how it differs	Search in a clean investigation profile, signed out (Chapter 8)
Localization	Adjusts results based on apparent location	A subject's local coverage may be invisible from elsewhere, and vice versa	Set the engine's region and language settings explicitly and record them
Language detection	Prioritizes results in your inferred language	Non-English material is systematically underweighted for English-locale users	Query in the target language; set language preferences; use regional engines
Freshness bias	Boosts recent content	Historical material is buried; a story's original reporting sits below its aggregations	Date-range filters; archives; explicit historical queries
Device signals	Alters results for mobile versus desktop	Different result sets, and mobile often shows fewer	Use desktop for investigative work; record which you used

All five are reasons to record your search conditions, and all five are reasons that a colleague repeating your search may see something different.

1.1.6 Verifying What This Chapter Says

Because the landscape changes, adopt this habit now:

1. **Check the engine's own about or documentation pages** for statements about its index and partnerships.
2. **Run a comparison query** across the engines you intend to use, on an obscure term, and observe whether the result sets are near-identical.
3. **Check the operator syntax** in the engine's current help documentation before relying on it, because operators are deprecated without notice.
4. **Note the date of your verification** in your search log, because it will change.

When this chapter and reality disagree, reality is correct. Record the difference and move on.

1.2 Search Method

1.2.1 Query Construction

Search skill is mostly query construction, and query construction is mostly discipline.

Principle	Explanation
Start narrow, then widen	A specific query that returns nothing is more informative than a broad one that returns everything, because you learn where the boundary is
Use the terms the source would use	Search the language of the document you want, not the language of your question. A filing says "notice of violation," not "got in trouble"
One variable at a time	Change one element per query so you can attribute the difference in results
Exact phrases for distinctive strings	Names, error messages, document titles, and stock phrases are the highest-yield queries in investigative work
Search for the artifact, not the topic	"Annual report" plus an entity name finds documents; "company financial information" finds articles about companies
Anticipate the page you want	Imagine the page that would answer your question, then search for words that would appear on it
Iterate and record	Every query is a data point about the index, and the log is what makes the iteration systematic

1.2.2 Operators, Conceptually

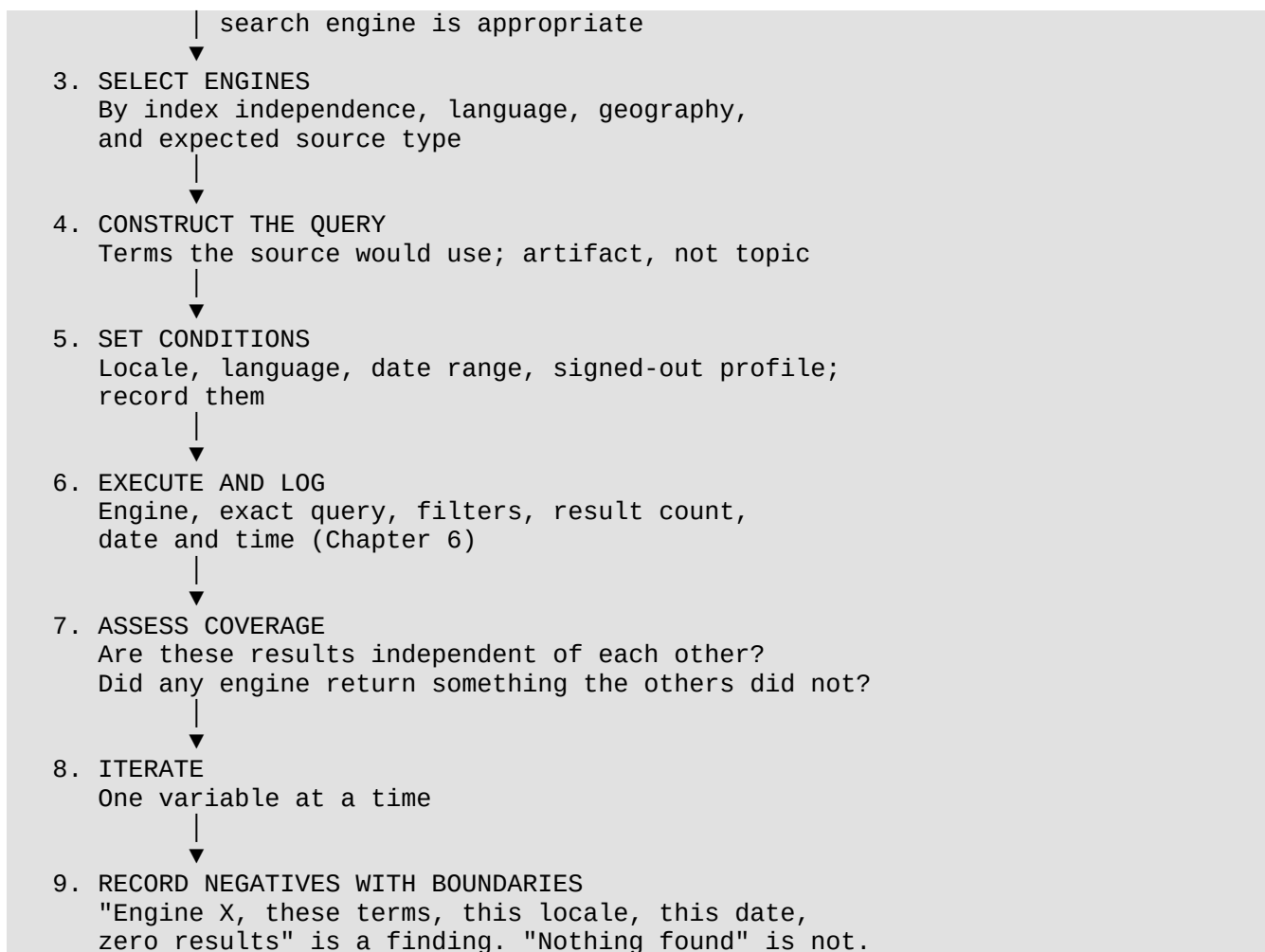
Every major engine supports some version of the following. **The specific syntax varies by engine and changes without notice**, so learn the concepts and verify the current syntax in each engine's documentation.

Concept	What it does	Investigative use
Exact phrase	Requires the words in that order	Names, titles, error strings, boilerplate language
Exclusion	Removes results containing a term	Cutting out a dominant unrelated meaning
Site restriction	Limits to one domain or subdomain	Searching a specific organization's own site, or a whole top-level domain
File type restriction	Limits to a document format	Finding filings, reports, presentations, and spreadsheets rather than pages
Title, URL, or body restriction	Requires a term to appear in a specific part of the page	Precision; finding directory pages and index listings
Date range	Limits by time	Historical work; excluding recent noise
Wildcards	Placeholder for an unknown term	Phrase variants
Numeric range	Matches a span of numbers	Prices, years, quantities
Related or similar	Finds comparable sites	Discovering an industry's population
Cache or archived view	Retrieves the engine's stored copy	Content removed or changed since crawling; increasingly unreliable and being withdrawn by several engines

Two cautions. First, operator support is uneven: an engine may accept an operator and ignore it, producing results that look filtered and are not. Test each operator on a query where you know the answer. Second, **operators do not expand the index**. A perfectly constructed query cannot return a page the crawler never fetched.

1.2.3 The Investigative Search Workflow

1. STATE THE QUESTION
What page, document, or fact would answer it?
↓
2. DECIDE: IS A SEARCH ENGINE THE RIGHT TOOL?
If the answer lives in a structured system (registry, docket, permit database), go there instead. (Chapter 3)



Step 2 is the step that separates competent investigators from busy ones. A great many questions that people put to a search engine are answered definitively by a structured system, and the search engine will give a worse answer more slowly.

1.2.4 Documenting Searches

Carried from Chapter 6 and specialized here. Every search log entry:

Field	Reason
Search ID	Referenced from findings
Engine and interface	Results differ by engine and by regional interface
Exact query string	Reproducibility, to the extent it exists
Operators used, and whether they appeared to work	Operators are silently ignored
Locale and language settings	The largest single cause of different results
Date range filter, if any	
Signed-in state and profile	Personalization

Field	Reason
Date and time with time zone	Indexes change
Result count as reported	An approximation, and still a data point
What was found, by item ID	Links to the evidence log
What was not found	With the coverage boundary
Screenshot of the results page	Because the search is not reproducible

The screenshot requirement is specific to this chapter. A search result set is ephemeral in a way a filing is not. If a result matters, capture the results page as well as the destination page.

1.2.5 OPSEC for Searching

Chapter 8 applies directly. Three points specific to search:

Your queries are observable. By the engine, by your network operator in some configurations, and by your organization. A query is a statement of interest.

Clicking through is a visit. The subject's server sees the visit and, in some configurations, the referring search. Chapter 8's separate profile and archive-first discipline apply.

Some engines are more revealing than others, both in what they retain and in what they pass to destinations. Section 1.6 and comparison table 2.6 cover this.

1.3 Google for OSINT

1.3.1 What It Is

The largest general-purpose index, with the most sophisticated ranking, the broadest feature set, and the most aggressive personalization and localization. For most investigators it is the default starting point, and the default is worth interrogating.

1.3.2 Strengths

Strength	Investigative value
Index breadth	The largest crawl of the general web; the highest probability that a page exists in the index
Operator support	The richest and most reliable operator set, including site, file type, title, URL, and date restriction
File type discovery	Exceptionally strong for finding documents rather than pages, which is where investigative value concentrates
Freshness	Fast recrawling of active sites

Strength	Investigative value
Ancillary indexes	Separate corpora for books, scholarly literature, patents, and news, each with its own search interface
Language handling	Strong across major languages, with translation integration
Feature richness	Answer boxes, site links, and structured presentations that expose page structure

1.3.3 Limitations

Limitation	Consequence
Heavy personalization	You do not know how your results differ from a neutral set
Heavy localization	Results shift substantially by apparent location
Freshness bias	Recent content outranks original historical reporting
Result depth restriction	Reaching deep into a result set is increasingly difficult, and reported counts are approximations
Feature crowding	Answer boxes and features displace organic results, particularly on mobile
Commercial pressure on results	Heavily monetized query spaces are dominated by optimized content
Operator deprecation	Operators have been quietly weakened or removed over time; verify before relying
Privacy	The most extensive data collection of the engines covered here

1.3.4 Investigative Techniques

Technique	Approach
Document hunting	Combine a file type restriction with an organization's domain or name to surface reports, filings, presentations, and spreadsheets
Site mapping	Restrict to a domain and iterate through sections to enumerate what an organization publishes
Top-level domain sweeps	Restrict to a country or sector domain to find institutional material
Exact-phrase provenance	Search a distinctive sentence to find where else it appears, which detects copied text and shared

Technique	Approach
	templates (Chapter 5)
Directory discovery	Search for terms characteristic of index listings to find published file directories
Historical narrowing	Combine date restriction with an exact phrase to find original reporting rather than aggregation
Ancillary corpora	Use the scholarly, patent, book, and news interfaces, which have different coverage from the main index

A note on so-called advanced searching. Collections of operator combinations circulate widely, and many of them are aimed at discovering exposed systems and files. Chapter 7 and Rule 2 govern here: finding material that was exposed by error does not make it public, and probing for exposed systems you do not own is outside this course.

1.3.5 Best Use Cases

Document discovery, broad initial coverage, English-language and major-language material, scholarly and patent work through the ancillary corpora, and any question where index breadth matters most.

Weakest for: material in a language or region it underweights, historical material buried by freshness bias, anything requiring an unpersonalized view, and anything in a structured system it cannot crawl.

1.4 Bing for OSINT

1.4.1 What It Is

The second major independent general index, and structurally important beyond its own user base because **a number of other engines have historically drawn on it**, which means Bing's coverage propagates into interfaces that appear independent.

1.4.2 Strengths

Strength	Investigative value
A genuinely independent index	Real second-opinion value against Google
Different ranking emphasis	Surfaces pages Google ranks low, particularly older and less optimized sites
Image search capability	Historically strong, with useful visual matching features
Operator support	A comparable set to Google's, with some differences worth learning
Less aggressive result-depth restriction	Historically easier to page deeper into results

Strength	Investigative value
Regional interfaces	Country-specific versions that change results meaningfully

1.4.3 Limitations

Limitation	Consequence
Smaller index than Google	Higher chance a page is absent
Weaker in some languages and regions	Notably outside its strongest markets
Personalization and localization	Present, though generally less pronounced
Feature crowding	Increasing, as with all major engines
Downstream dependency	Because other interfaces have drawn on it, "checking Bing and two others" may be checking Bing three times

1.4.4 Investigative Techniques

- **Use it as the primary independence check against Google.** If a page appears in one and not the other, that is a finding about coverage.
- **Use it for older and less optimized sites**, which its ranking treats more generously.
- **Use its image search** as a second visual-matching opinion (Section 1.7 covers the strongest option).
- **Use regional interfaces** deliberately, and record which one you used.

1.4.5 Best Use Cases

Independence checking, older web content, image work, and any investigation where two independent general indexes are needed for corroboration.

1.5 DuckDuckGo for OSINT

1.5.1 What It Is

A privacy-focused search interface. Its distinguishing characteristic is what it does not do: it does not build a profile of the user, does not personalize results on the basis of search history, and does not pass identifying information to destinations in the way personalized engines do.

On its index: DuckDuckGo has historically drawn its general web results substantially from a partner index while operating its own crawler for some purposes and combining multiple sources. **Treat the current arrangement as something to verify** (Section 1.1.6). The investigative consequence is the

important part: if the results derive substantially from another engine's index, DuckDuckGo is a privacy improvement rather than a coverage improvement.

1.5.2 Strengths

Strength	Investigative value
No personalization	Every user sees the same results for the same query and settings, which makes results comparable between investigators and across time
No search history profiling	Your investigative interests are not accumulated into a profile
Reduced referrer leakage	Less information passed to destination sites about your query
Region and time filters	Straightforward, with an explicit region selector
Direct-jump shortcuts	Query prefixes that send a search to another site, which is a workflow convenience
Consistent, uncluttered results	Easier to compare systematically

1.5.3 Limitations

Limitation	Consequence
Coverage may not be independent	Agreement with a partner engine is not corroboration
Smaller effective index than Google	Pages missing
Weaker operator support	Some operators unsupported or inconsistently applied
Weaker in specialized corpora	No equivalent to the scholarly and patent indexes
Privacy is not anonymity	Chapter 8 applies: the network path and your browser are unchanged

1.5.4 Investigative Techniques

- **Use it as the baseline for reproducible comparison**, because the absence of personalization makes results comparable across investigators, which is exactly what Lab 1 needs.
- **Use the region selector explicitly** and record it.
- **Use it when the query itself is sensitive**, which is a genuine consideration when researching a subject who could obtain search data.
- **Do not treat it as an independence check** against its partner engine.

1.5.5 Best Use Cases

Privacy-sensitive queries, reproducible comparison work, and general searching where personalization would distort the result.

1.6 Brave Search for OSINT

1.6.1 What It Is

A search engine that has built **its own independent index**, which is unusual and is the reason it belongs in this chapter as a distinct entry rather than as another privacy interface.

On its index: Brave has publicly stated that it operates an independent index built partly from a community contribution mechanism, with the ability to fall back to third-party results in some circumstances, and it has published an indicator of how much of a given result set came from its own index. **Verify the current arrangement** (Section 1.1.6). The presence of an independence indicator, where it exists, is directly useful to an investigator.

1.6.2 Strengths

Strength	Investigative value
Independent index	Genuine coverage difference from both Google and Bing
Independence transparency	Where an indicator is provided, you can see how much of a result set is its own
No personalization profiling	Comparable results across users
Different ranking	Surfaces pages the majors rank low
Privacy posture	Comparable in intent to DuckDuckGo

1.6.3 Limitations

Limitation	Consequence
Smaller index	A younger crawl than the majors, with real gaps
Weaker in non-English and regional material	Coverage thins outside its strongest areas
Operator support less mature	Verify each operator
Fallback behaviour	Some results may come from a third party, which affects independence claims
Less depth in specialized content	No scholarly or patent equivalents

1.6.4 Investigative Techniques

- **Use it as a third independent index**, alongside Google and Bing, which is the strongest reason to include it.
- **Watch the independence indicator** where available and record it, because it tells you whether a result set is a genuine second opinion.
- **Use it for obscure and less commercially valuable material**, where a different crawler's priorities show.

1.6.5 Best Use Cases

Independence checking as a third opinion, privacy-conscious searching where coverage also matters, and finding material the majors rank into invisibility.

1.7 Yandex for OSINT

1.7.1 What It Is

A major independent index with its origins and strongest coverage in Russian-language and regional material, and with image-matching capability that investigators have long regarded as exceptional.

1.7.2 Strengths

Strength	Investigative value
Reverse image search	Widely regarded as the strongest general-purpose visual matching available, particularly for faces, buildings, landscapes, and cropped or altered images, and notably better than the majors at finding visually similar rather than identical images
Independent index	Genuine coverage difference
Russian-language and regional coverage	Substantially better than global engines for that material
Operator support	A distinct set, including some capabilities the majors lack
Different ranking priorities	Surfaces material the majors do not

1.7.3 Limitations and Considerations

Limitation	Consequence
Weaker coverage outside its core regions	Not a substitute for the majors on Western material

Limitation	Consequence
Interface and operator differences	A learning curve, and documentation may be in Russian
Accessibility	Availability varies; some organizations and jurisdictions restrict access
Organizational policy	Many employers have policies on which foreign services may be used; check before use
Data handling	Your queries are processed under a different legal regime, which is an OPSEC and a compliance consideration (Chapters 7 and 8)

This course's position. Yandex's reverse image capability is genuinely valuable and is used widely in open source investigation, including in well-known public-interest work. Whether you may use it is a question for your organization's policy and your own threat model, not one this chapter can answer for you. **Where policy prohibits it, the lab paths in Part 8 do not require it.**

1.7.4 Investigative Techniques

- **Reverse image search is the primary reason to use it.** For geolocation work (Chapter 2), it frequently returns visually similar scenes that identical-match engines miss.
- **Use it for Russian-language and regional material**, with queries in the target language.
- **Use it as an independence check** on general web results, recognizing its regional weighting.

1.7.5 Best Use Cases

Reverse image search and visual similarity matching; Russian-language and regional research; independence checking with a regional bias.

1.8 Baidu for OSINT

1.8.1 What It Is

The dominant search engine in mainland China, with a large independent index concentrated in Chinese-language content.

1.8.2 Strengths

Strength	Investigative value
Chinese-language coverage	The deepest index of Chinese-language web content, substantially exceeding what global engines hold
Domestic Chinese sites	Indexes sites that global crawlers reach poorly or

Strength	Investigative value
	not at all
Independent index	Complete coverage difference from the Western majors
Ancillary services	Encyclopedic, mapping, and document-hosting services with their own content

1.8.3 Limitations and Considerations

Limitation	Consequence
Chinese-language interface and queries	Effective use requires the language, or careful use of translation
Content moderation	Results are subject to regulatory filtering, which is itself an investigative finding when observed
Weak coverage outside its core	Not useful for non-Chinese material
Accessibility and policy	As with Yandex: availability and organizational policy vary, and queries are processed under a different legal regime
Result verification	Results should be verified against other sources where possible, as with any single index

1.8.4 Investigative Techniques

- **Query in Chinese.** Searching in English returns very little of the value. Use translated terms and, where possible, have the translation checked, because term choice matters enormously.
- **Compare with global engines deliberately.** The difference between what Baidu returns and what Google returns for the same Chinese-language query is itself informative.
- **Use it for corporate and supply chain research** involving Chinese entities, where domestic sources hold material global engines lack.
- **Treat filtering as data.** A topic that returns substantive results elsewhere and nothing here is a finding about the information environment, not about the topic.

1.8.5 Best Use Cases

Chinese-language corporate, supply chain, and technical research; understanding the domestic Chinese information environment; any investigation involving entities operating primarily in mainland China.

1.9 Mojeek for OSINT

1.9.1 What It Is

A small independent search engine that operates **its own crawler and index**, with no personalization and no result reordering based on the user. It is included in this chapter not because it is large but because it is genuinely independent, which is rare.

1.9.2 Strengths

Strength	Investigative value
Fully independent index	A real and unusual third or fourth opinion
No personalization	Perfectly comparable results across users and sessions
Different crawl priorities	Indexes small, old, and low-traffic sites that the majors deprioritize
Stable, uncluttered results	Easy to compare systematically
Date-based sorting	Useful for freshness and historical work
Privacy posture	No tracking

1.9.3 Limitations

Limitation	Consequence
Much smaller index	Many pages simply absent
Weaker ranking	Relevance is less refined than the majors
Limited non-English coverage	Thin outside English
Few features	No specialized corpora or rich result features

1.9.4 Investigative Techniques

- **Use it precisely because it is small and different.** Its crawler reaches long-tail sites the majors ignore, which is where obscure organizational material frequently lives.
- **Use it as an independence check** where you need to know that a result is not derived from a major index.
- **Use it for historical and small-site work**, including hobbyist, community, and legacy pages.
- **Do not expect it to answer mainstream queries well.** That is not what it is for.

1.9.5 Best Use Cases

Independence verification, long-tail and small-site discovery, historical web material, and any situation where knowing that a result did not come from a major index matters.

1.10 Regional Search Engines for OSINT

1.10.1 Why Regional Engines Matter

Global engines are optimized for global queries and majority languages. Regional engines are frequently better at:

Advantage	Explanation
Local-language content	Deeper indexes of material in the national language, including morphology and search conventions the global engines handle poorly
Local sites	Domestic sites, forums, directories, and portals that global crawlers reach shallowly
Local conventions	Address formats, business identifiers, and naming conventions handled natively
Local services	Integrated mapping, business directories, and reference content with local depth
Cultural context	Query interpretation shaped by local usage

The investigative consequence: for a question about an entity, a place, or an event in a particular country, the national engine frequently returns material that global engines never surface, and sometimes never indexed at all.

1.10.2 A Survey

This list is illustrative and it will age. Engines close, merge, and change their index arrangements. Verify before relying, and check whether an engine has an independent index or draws on a partner.

Region	Engine type	Notes for investigators
South Korea	A dominant national portal-search hybrid	Much of the Korean web lives inside portal ecosystems (blogs, cafés, knowledge services) that global engines index poorly; the national engine reaches inside them
Czechia	A long-established national engine with its own index	Strong for Czech-language and domestic content
Japan	Global engines dominate, with a major portal that has historically been powered by one of them	Portal-hosted content (auctions, knowledge boards, blogs) matters more than the search engine choice
France	A privacy-focused national	Useful for French-language and

Region	Engine type	Notes for investigators
	engine, historically hybrid in its index sourcing	EU-focused material; verify index sourcing
Germany and Europe	Several privacy-focused and environmentally themed engines, mostly drawing on partner indexes	Privacy value; limited independent coverage value
Russia and the region	Covered in Section 1.7	
Mainland China	Covered in Section 1.8, plus other domestic engines with their own indexes	Multiple domestic engines exist; comparing them is informative
Vietnam	A domestic engine with local-language strength	Better for Vietnamese-language content
Middle East and North Africa	Regional portals and Arabic-language interfaces	Arabic morphology handling varies substantially; query construction matters more than usual
India	Global engines dominate; local-language coverage is uneven	Indic-language queries frequently underperform; regional news and government portals are the better route
Latin America	Global engines dominate	Country-level government and news portals matter more than any regional engine

A structural observation. In several markets, the important asset is not a national search engine but a **national portal ecosystem**: a platform whose blogs, forums, and knowledge services hold enormous amounts of content that general crawlers cannot see well. In those markets, the right move is often to search inside the portal rather than to find a different engine.

1.10.3 Working With a Regional Engine You Do Not Read

A practical procedure:

1. **Translate your query terms carefully**, and prefer terms taken from a document in the target language over machine-translated phrasing. Search the words the source would use.
2. **Try several term variants**, because translation choices change results dramatically.
3. **Use the engine's own interface**, not a translated proxy, so that ranking and features work normally.
4. **Translate the results page** for triage, then work with the original for anything you rely on.
5. **Have anything material checked by a speaker.** Machine translation is adequate for triage and

inadequate for a finding.

6. **Record the original-language terms** in your search log, not just the English.
7. **Note the transliteration problem:** names and entities may appear in several romanizations, and each is a separate query.

1.10.4 Legal, Policy, and OPSEC Considerations

Consideration	Guidance
Organizational policy	Many employers restrict which foreign services may be used; check before, not after
Data handling	Your queries are processed under another jurisdiction's regime (Chapter 7)
Accessibility	Some engines are unreachable from some networks or jurisdictions
Attribution	Using a regional engine from an organizational network is itself a signal (Chapter 8)
Terms of service	Automated querying is prohibited on all of them (Rule 5)
Content filtering	Results may be filtered by regulation, which affects your coverage boundary and is itself a finding

1.11 The Multi-Engine Method

1.11.1 Why Multiple Engines

Four reasons, in order of importance:

Coverage. Independent indexes hold different pages. Using one engine means accepting one crawler's priorities as the boundary of the web.

Corroboration. Two independent indexes returning the same page is weak corroboration of the page's existence and prominence; it says nothing about the content's truth. Two engines returning the same page because they share an index says nothing at all.

Ranking diversity. The same index ranked differently exposes different material. A page that is result 80 on one engine may be result 4 on another.

Failure detection. When one engine returns nothing and another returns a great deal, you have learned that the first engine's silence was about its index, which is Rule 10 made operational.

1.11.2 A Working Sequence

1. ONE MAJOR INDEX FIRST
Establish the baseline and the obvious material
↓
2. THE SECOND MAJOR INDEX
Independence check; note what appears in one and not the other
↓
3. AN INDEPENDENT SMALLER INDEX
Long-tail and low-traffic material
↓
4. A REGIONAL ENGINE, IF THE SUBJECT HAS A REGION
Native-language queries
↓
5. SPECIALIZED CORPORA
Scholarly, patent, news, books, as the question requires
↓
6. ARCHIVES
For anything removed or changed (Chapter 3)
↓
7. STRUCTURED SOURCES
Registries, dockets, permits: the answer may have been there the whole time (Chapter 3)
↓
8. RECORD COVERAGE
Which engines, which queries, which locales, what each uniquely contributed, and what none of them returned

Step 7 is placed last deliberately and should often be first. The sequence above is for questions that genuinely require web search. Many questions do not.

1.11.3 Diminishing Returns

Multi-engine searching has a stopping point, and Chapter 4's saturation concept applies:

Signal	Interpretation
The third engine returns nothing the first two did not	Coverage saturation for this query; change the query, not the engine
Every engine returns the same top results	Either the material is genuinely concentrated, or the engines share an index
A regional engine returns a great deal the others did not	You have found a coverage gap; keep working that seam
All engines return nothing	Either the material is not on the indexed web, or your query is wrong. Change one variable and test

1.11.4 Recording Coverage

The output of multi-engine searching is a coverage statement, and it belongs in every product that rests on web search:

"Searched Google, Bing, Brave, and Mojeek between 4 and 6 June, using the term sets recorded in the search log, with locale set to [X] and language to [Y], signed out. Google and Bing returned overlapping results; Brave returned two items absent from both; Mojeek returned one archived community page absent from all. No engine returned any result matching [specific query], which does not establish that no such material exists, only that it is not indexed by these four engines under these terms as of the search date."

That last sentence is Rule 10 written out, and it belongs in products more often than it appears.

Part 2: Comparison Tables

A standing caveat for this Part. These tables describe the landscape as understood at the time of writing. Index arrangements, partnerships, operator support, and feature sets change without notice. Verify anything you intend to rely on, using the method in Section 1.1.6, and record the date of your verification.

2.1 Search Engines vs. Metasearch Engines

Dimension	Search engine	Metasearch engine	Privacy proxy
Runs its own crawler	Yes, if independently indexed	No	No
Holds its own index	Yes	No	No
Source of results	Its own index	Several engines, merged	One engine, relayed
Ranking	Its own	A merge of others' rankings, frequently crude	The source engine's
Coverage benefit	Genuine, if independent	Breadth of presentation, not of index	None
Privacy benefit	Varies	Depends on implementation	The primary purpose
Operator support	Full, per engine	Inconsistent; operators may not pass through	Usually passes through
Freshness	Its own crawl cadence	The slowest contributor	The source engine's
Investigative use	Primary tool	Quick breadth pass; convenience	Sensitive queries against a major index
Investigative caution	Know whether the index is independent	Merged results obscure which engine found what	Coverage is unchanged; only exposure changes

Dimension	Search engine	Metasearch engine	Privacy proxy
Failure mode	Index gaps	Attributing a finding to the wrong engine	Believing privacy equals different coverage

2.2 Google vs. Bing

Dimension	Google	Bing
Index size	Largest	Large, smaller than Google
Independence	Independent	Independent
Ranking emphasis	Authority, freshness, engagement signals	Somewhat more generous to older and less optimized sites
Operator support	Richest and most reliable	Comparable set, different specifics
File type discovery	Excellent	Good
Specialized corpora	Scholarly, patents, books, news	News; fewer specialized corpora
Image search	Good, identical-match oriented	Strong, with useful visual matching features
Personalization	Heavy	Present, generally lighter
Localization	Heavy	Present
Result depth	Increasingly restricted	Historically easier to page deeper
Regional interfaces	Extensive	Extensive
Downstream use by other engines	Some proxies relay it	Historically a source for several interfaces
Best investigative use	Document discovery, breadth, specialized corpora	Independence check, older content, image work
Chief limitation	Personalization opacity and freshness bias	Smaller index; downstream dependency inflates apparent independence

2.3 DuckDuckGo vs. Brave

Dimension	DuckDuckGo	Brave Search
Own index	Partial at most; results substantially sourced from a partner historically	Yes, an independent index
Independence value	Low, if it shares a partner's index	High
Independence transparency	Not surfaced to the user	An indicator of own-index

Dimension	DuckDuckGo	Brave Search
		proportion has been provided
Personalization	None	None
Result reproducibility across users	High	High
Privacy posture	Strong; no search profiling	Strong; no search profiling
Operator support	Moderate, inconsistent	Moderate, less mature
Region and language controls	Explicit region selector	Present
Specialized corpora	None	None
Coverage of long-tail sites	Inherits the partner's	Different, sometimes better, sometimes worse
Workflow features	Direct-jump shortcuts to other sites	Fewer
Best investigative use	Reproducible comparison baseline; sensitive queries	Third independent index; finding what the majors bury
Chief limitation	Coverage is not independent	Smaller and younger index; weaker outside English

2.4 Independent Indexes vs. Aggregated Results

Dimension	Independent index	Aggregated, licensed, or proxied
Definition	The operator crawls and indexes the web itself	The operator presents another's results
Examples of the type	Google, Bing, Yandex, Baidu, Mojeek, Brave	Metasearch tools, privacy proxies, and interfaces licensing a partner index
Coverage difference from the source	Genuine	None
Corroboration value	Real: agreement across independent indexes means two crawlers found it	None: agreement is one index seen twice
Ranking difference	Yes	Sometimes, if the aggregator re-ranks
Freshness	Its own crawl cadence	The source's
Operator support	Its own	Whatever passes through
How to identify	Published crawler documentation; distinctive results on obscure queries	Near-identical result sets to a known engine

Dimension	Independent index	Aggregated, licensed, or proxied
Investigative use	Coverage and corroboration	Privacy, convenience, or presentation
Failure mode	Assuming a large index is complete	Counting one index as several sources

2.5 Global vs. Regional Search Engines

Dimension	Global engines	Regional engines
Index priority	Major languages, high-traffic sites, global commerce	National language, domestic sites, local services
Local-language depth	Variable; frequently shallow	Deep
Domestic site coverage	Shallow crawl of many national sites	Thorough
Local conventions	Poorly handled: address formats, identifiers, naming	Native
Portal ecosystems	Poorly indexed	Frequently integrated
Query interpretation	Global usage	Local usage
Interface language	Yours	Frequently the local language
Accessibility	Generally available	Varies; some restricted or unreachable
Regulatory filtering	Varies by market	Frequently significant, and itself informative
Investigative use	Baseline and breadth	Reaching material global engines never indexed
Failure mode	Concluding material does not exist because global engines lack it	Assuming the regional engine is complete for non-local material

2.6 Privacy Features

Engine	Search history profiling	Personalization	Data sharing with destinations	Notes for investigators
Google	Extensive, especially when signed in	Heavy	Substantial	Use signed out in an investigation profile; assume the query is retained
Bing	Present	Present	Present	Similar posture;

Engine	Search history profiling	Personalization	Data sharing with destinations	Notes for investigators
				regional interfaces vary
DuckDuckGo	None	None	Reduced	Strongest for sensitive queries; privacy is not anonymity
Brave Search	None	None	Reduced	Privacy plus independent coverage
Yandex	Present	Present	Present	Processed under a different legal regime; policy question
Baidu	Present	Present	Present	Processed under a different legal regime; policy question
Mojeek	None	None	Reduced	No tracking; small index
Metasearch tools	Varies by implementation	Usually none	Varies	Self-hosted instances can be strong; public instances vary
Privacy proxies	None at the proxy	The source engine's, without your identity	Reduced	Coverage is the source engine's

Two cautions. Privacy is not anonymity: none of these changes your IP address, your browser fingerprint, or your organization's visibility (Chapter 8). And a privacy-preserving engine drawing on a partner index still sends your query to that partner in some architectures, which is a question worth verifying.

2.7 Image Search Capabilities

Engine	Identical-image matching	Visually similar matching	Face-related matching	Investigative strength
Yandex	Good	Strongest of the general engines	Notably capable	Geolocation, cropped and altered images, finding scenes rather than files

Engine	Identical-image matching	Visually similar matching	Face-related matching	Investigative strength
Google	Strong	Moderate	Restricted	Finding the same file elsewhere; object and text recognition in images
Bing	Strong	Good	Restricted	Useful second opinion; visual matching features
Brave	Present	Limited	No	Limited
DuckDuckGo	Present, sourced	Limited	No	Limited
Baidu	Good within its corpus	Good	Present	Chinese-language and domestic content
Mojeek	Limited	Limited	No	Not an image tool

Method note. Chapter 2's verification sequence applies: search the whole image, then cropped regions separately, because composites and altered images frequently fail whole-image matching. Different engines succeed on different crops, which is a reason to use several.

2.8 Language Coverage

Engine	English	Major European	Russian and regional	Chinese	Japanese and Korean	Arabic	Indic and other
Google	Excellent	Excellent	Good	Moderate	Good	Moderate	Variable, frequently weak
Bing	Excellent	Good	Moderate	Limited	Moderate	Moderate	Weak
DuckDuckGo	Good	Good	Limited	Limited	Limited	Limited	Weak
Brave	Good	Moderate	Limited	Limited	Limited	Limited	Weak
Yandex	Moderate	Moderate	Excellent	Limited	Limited	Limited	Weak
Baidu	Limited	Limited	Limited	Excellent	Limited	Limited	Weak
Mojeek	Good	Moderate	Weak	Weak	Weak	Weak	Weak
National engines	Varies	Varies			Strong in their market	Varies	Varies

The practical rule: for any investigation whose subject operates primarily in a non-English language, the global engines are a starting point and not a coverage claim. Query in the target language and add a regional engine.

2.9 Regional Strengths

Region	Best coverage from	Why	What global engines miss
North America, UK, Ireland, Australia	Google, Bing	Deep crawls, mature ranking	Little; the strongest global coverage
Western Europe	Google, plus national engines	Global depth plus national-language depth	Local directories, regional press, national portals
Central and Eastern Europe	Google plus national engines	National engines index domestic content more deeply	Domestic forums, national-language press
Russia and neighbouring markets	Yandex	Native index and language handling	Domestic sites, regional press, portal content
Mainland China	Baidu and other domestic engines	Domestic crawl and regulatory environment	The large majority of the domestic Chinese web
Japan	Global engines, plus portal ecosystems	Portals hold the content	Auction, knowledge-board, and blog ecosystems
South Korea	The national portal-search hybrid	Content lives inside the portal	Blogs, cafés, and knowledge services
Southeast Asia	Global engines plus domestic engines	Mixed	Local-language forums and domestic portals
Middle East and North Africa	Global engines with Arabic queries, plus regional portals	Morphology handling varies	Local press and forums
Latin America	Global engines with local-language queries	Global coverage is reasonable	Government portals and local press are better reached directly
Sub-Saharan Africa	Global engines	Limited alternatives	Substantial material is not on the indexed web at all

2.10 Advantages, Limitations, and Best OSINT Use Cases

Engine	Advantages	Limitations	Best use	Avoid for
Google	Largest index; best	Personalization;	Document	Unpersonalized

Engine	Advantages	Limitations	Best use	Avoid for
	operators; file type discovery; specialized corpora	freshness bias; restricted depth; heaviest data collection	discovery, breadth, scholarly and patent work	baselines; non-major languages; historical depth
Bing	Independent index; older content; image features; deeper paging	Smaller; downstream dependency inflates apparent independence	Independence checking; older web; image work	Assuming it plus its downstream users are three sources
DuckDuckGo	No personalization; reproducible; privacy; region control	Coverage may not be independent; weaker operators	Reproducible comparison; sensitive queries	Independence checking against its partner
Brave	Independent index; independence indicator; privacy	Smaller, younger; weaker outside English	Third independent opinion; material the majors bury	Non-English and regional depth
Yandex	Best-in-class visual similarity matching; independent index; regional depth	Weak outside its regions; policy and access considerations	Reverse image and geolocation work; Russian-language research	Western-market general web as a primary source
Baidu	Deepest Chinese-language index; domestic sites	Requires Chinese; filtering; policy and access considerations	Chinese entity, supply chain, and technical research	Anything outside Chinese-language content
Mojeek	Fully independent; no personalization; long-tail crawl	Small; weaker ranking; English-centric	Independence verification; small and old sites	Mainstream queries; non-English
Regional engines	National-language and domestic depth; local conventions	Narrow; interface language; variable access	Country-specific investigations	Anything outside their market
Metasearch	Breadth in one pass	No index; attribution of findings unclear; operators unreliable	Quick initial sweep	Coverage claims and corroboration

Part 3: Case Studies

Each case explains why particular engines were selected, what each returned, what was unique to one engine, and how the combination improved the investigation. Cases marked *composite* are built from documented professional practice and are not real organizations or people. Engine behaviour described reflects the period in which the cases were written.

3.1 Cybersecurity: Identifying a Vulnerable Software Component

(*composite*)

The situation. A security team learns that a component used in one of its products has a newly disclosed flaw. They need the vendor advisory, the affected version range, any public exploitation reporting, and the component's presence in their own supplier ecosystem, within a day.

Engine selection and reasoning.

Engine	Why selected	What it returned
Google	Breadth and file type discovery for vendor PDFs	The vendor advisory, three vendor PDFs with affected version tables, and the bulk of the news coverage
Bing	Independence check; historically better paging depth	Two vendor bulletins from smaller downstream vendors that Google ranked below the visible range
Mojeek	Long-tail crawl for small sites	A mailing list archive on a small domain, discussing the same behaviour eighteen months earlier
Brave	Third independent index	Overlapped heavily with Google and Bing; one forum thread unique
A regional engine	The component's original developer is in a non-English-speaking country	The developer's own release notes in the original language, more detailed than the English advisory

Unique findings. Three items came from exactly one engine each: the downstream vendor bulletins (Bing), the eighteen-month-old mailing list discussion (Mojeek), and the original-language release notes (regional engine). The mailing list item changed the assessment materially, because it indicated the behaviour had been observed and discussed long before the formal disclosure.

How the combination helped. The English advisory gave a version range. The original-language

release notes gave a more precise range, and the two did not agree. The team reported the discrepancy rather than choosing one, which is Chapter 5's discipline applied to a search finding.

Lessons.

The small independent index earned its place, and it did so exactly where predicted: an old page on a low-traffic domain.

Original-language sources are frequently more precise than their translations, and a single-engine, single-language search would have produced a confidently wrong version range.

Paging depth is an engine characteristic worth knowing. The Bing findings were not absent from Google's index; they were below where anyone looks.

3.2 Corporate Research: Establishing an Entity's Real Footprint

(composite)

The situation. An analyst must establish where a mid-sized manufacturer actually operates, against a website that lists four locations.

Engine selection and reasoning.

Engine	Why selected	What it returned
Google	Document discovery via file type restriction on the company's domain and name	Two investor presentations and a supplier quality manual listing six sites
Bing	Independence check	A cached-era trade directory listing with a seventh address
Mojeek	Long-tail	An old local chamber of commerce page naming a site the company no longer mentions
Regional engine (the company's second market)	Local-language coverage	Local press coverage of a plant opening and a subsequent dispute, neither in English
Archives (not an engine, per Chapter 3)	Prior versions of the locations page	The locations page listed seven sites two years earlier

Unique findings. The seventh address (Bing), the disused site (Mojeek), the local dispute (regional engine), and the historical locations page (archives). Four sources, four distinct mechanisms.

How the combination helped. The website's four locations became a documented seven, with two no longer operating and one never publicly acknowledged in English. The analyst then went to structured sources (Chapter 3): property records and permits for each address, which is where the actual confirmation came from.

Lessons.

Search engines found the addresses; structured records confirmed them. The engines were a discovery mechanism, not a source of proof.

The archives finding shows why they belong in every search plan. A removed page is invisible to every engine and available in an archive.

A regional engine plus a local-language query produced material that no amount of English querying would have surfaced.

3.3 Competitive Intelligence: Detecting an Expansion Before It Is Announced

(composite)

The situation. A firm needs to know whether a competitor is preparing to enter a regional market, ahead of its own investment decision.

Engine selection and reasoning.

Engine	Why selected	What it returned
Google	Breadth; job postings; file type discovery	Job postings for roles in the target region; an investor call transcript mentioning "adjacent geographies"
Bing	Independence; older content	A local business journal item about a lease, not indexed by Google at the time of searching
Brave	Independent third index	A regional trade association member listing showing a new entry
Regional engine	The target region's national engine	A municipal planning notice in the local language, published on a council site with no English version
Specialized corpora	Patents and scholarly indexes	Nothing relevant; recorded as a negative with the boundary

Unique findings. The lease item (Bing), the association listing (Brave), and the planning notice (regional engine). The planning notice was the decisive item and it named the company, the parcel, and the intended use.

How the combination helped. No single engine contained more than two of the four indicators. An investigator using only the largest index would have had job postings and a vague transcript, which is

suggestive and not decisive.

Lessons.

Municipal and planning material is frequently indexed only by national engines, because global crawlers reach local government sites shallowly.

The negative result from the specialized corpora was recorded with its boundary, which is what makes the rest of the coverage statement credible.

Four indicators from four engines is a stronger position than four indicators from one, because independent indexes are independent evidence of publication, which is a Chapter 5 point.

3.4 Investigative Journalism: Tracing the Origin of a Claim

(composite, reflecting documented newsroom practice)

The situation. A claim about a public body's spending is circulating widely. A reporter must establish where it originated and whether it is supported.

Engine selection and reasoning.

Engine	Why selected	What it returned
Google with date restriction	Freshness bias makes original reporting hard to find; date restriction counteracts it	Nine hundred recent items; restricted to a six-month window two years earlier, eleven items
Mojeek with date sorting	Independent index and date-based sorting	The earliest indexed appearance of the exact phrase, on a small advocacy site
Bing	Independence check on the earliest appearance	An earlier appearance still, on a regional news site
Exact-phrase searching across all engines	Provenance tracing (Chapter 5)	The identical sentence on fourteen sites, all after the two earliest
Archives	The advocacy site's earlier versions	The claim's first appearance, with a source citation that the current version has removed

Unique findings. The earliest indexed appearance (Bing), the removed source citation (archives), and the pattern of identical phrasing across fourteen sites, which is circular reporting made visible.

How the combination helped. The reporter established that fourteen apparently independent sources were one source repeated, that the original citation pointed to a document that did not say what was claimed, and that the citation had subsequently been removed from the originating page. The claim was reported as unsupported, with the provenance chain shown.

Lessons.

Exact-phrase searching across multiple engines is the primary tool for detecting circular reporting, and it is cheap.

Freshness bias actively obstructs provenance work, and date restriction is the counter.

Archives held the decisive item. The removed citation existed nowhere in any live index.

3.5 Academic Research: Building a Literature Base Across Languages

(composite)

The situation. A researcher needs the literature on a narrow engineering topic, including non-English work.

Engine selection and reasoning.

Engine	Why selected	What it returned
Scholarly corpus (Google)	Purpose-built for the task	The core English-language literature and citation chains
Google general web	Grey literature: technical reports, standards drafts, theses	Four institutional repository theses and two standards working documents
Bing	Independence	Two conference proceedings from a smaller publisher
Baidu with Chinese-language queries	A substantial body of work in the field is published in Chinese	Eleven items, of which four had no English abstract and were not in the scholarly index
Yandex with Russian-language queries	Similar reasoning	Six items, three unavailable elsewhere
Mojeek	Long-tail institutional pages	Two departmental pages listing unpublished technical reports

Unique findings. The Chinese and Russian-language work, which represented roughly a fifth of the relevant literature and was invisible to every English-language search.

How the combination helped. The researcher's literature review moved from "the field says X" to "the English-language literature says X; there is a body of work in two other languages that qualifies it," which is a materially different and more accurate claim.

Lessons.

Language coverage is the largest single blind spot in search-based research. A monolingual literature search produces a monolingual conclusion and rarely says so.

Grey literature lives on the general web, not in scholarly indexes, and file type restriction is how you find it.

Machine translation was adequate for triage and inadequate for citation, so anything relied upon was checked by a reader of the language, which is the Section 1.10.3 procedure.

3.6 Public Records Research: Finding the System, Not the Record

(composite)

The situation. An investigator needs property, permit, and licensing records for an address in a jurisdiction they have never worked in.

Engine selection and reasoning.

Engine	Why selected	What it returned
Google	Finding the systems	The county's assessor and recorder portals, the state licensing board, and a state open data portal
Bing	Independence	A municipal permit portal Google ranked below the visible range
Regional or local-language engine	Not required here; the jurisdiction is English-speaking	Not used, recorded as not applicable
Mojeek	Long-tail	An archived county page describing a records system that has since moved, with the historical coverage dates

The crucial finding. None of the actual records came from a search engine. **Every record came from querying the portals directly**, which is Chapter 3's deep web point. What the engines provided was the map: which systems exist, where they are, and what they cover.

How the combination helped. The Mojeek finding was disproportionately useful: an archived county page stating that records before a certain year were held in a different system and had not been migrated. That single fact defined the coverage boundary for the whole records search and would otherwise have produced a false negative.

Lessons.

Search engines find systems; systems hold records. Investigators who search for the record rather than for the system waste substantial time and reach worse answers.

Coverage statements about a records system are frequently published and rarely read. Finding one converts a guess about a negative result into a documented boundary.

A negative result from a records portal, with its published coverage dates attached, is a professional finding. The same negative without the dates is nothing.

3.7 International Investigation: One Entity, Four Information Environments

(composite)

The situation. A supply chain investigation concerns an entity with operations in four countries: an English-speaking market, a European market, Russia, and mainland China.

Engine selection and reasoning.

Environment	Engines used	Rationale	Result
English-speaking	Google, Bing, Brave	Breadth plus independence	Corporate filings, trade press, a court docket reference
European	Google plus the national engine, local-language queries	National engine indexes domestic press and registries more deeply	Local press coverage of a regulatory matter, absent from English results
Russia	Yandex, Russian-language queries	Native index and language handling	Registry-adjacent content and regional press
Mainland China	Baidu, Chinese-language queries	The domestic web is largely invisible to global engines	Domestic supplier directory listings and a trade association page naming the entity

Unique findings. In every environment except the English-speaking one, the majority of the useful material came from the non-global engine. The Chinese supplier directory listing named two subsidiaries that appeared in no Western source.

How the combination helped. The entity's structure as understood from English sources had three subsidiaries; the full picture had seven. Any assessment built on the English-language view would have been substantially wrong about the entity's footprint.

Complications the team recorded.

- **Policy:** the firm's policy required approval before using two of the engines, which was obtained and recorded.
- **Language:** all non-English findings were triaged with machine translation and verified by speakers before being relied upon.
- **Filtering:** one topic returned substantive results in three environments and nothing in the fourth, which was recorded as an observation about the information environment rather than as

a finding about the entity.

- **Transliteration:** the entity's name appeared in four romanizations, each of which was a separate query.

Lessons.

A single-language, single-engine investigation of a multinational entity is not an investigation of that entity. It is an investigation of what one index holds about it in one language.

Transliteration multiplies the query set, and missing a variant produces a false negative that looks like a finding.

Regulatory filtering is a coverage boundary and an observation, and it should appear in the product as both.

3.8 Cross-Case Comparison

Case	Decisive engine	Why it was decisive	What a single-engine investigation would have concluded
Vulnerable component	Mojeek and a regional engine	An old mailing list post and original-language release notes	A confidently wrong affected version range
Corporate footprint	Regional engine and archives	Local press and a removed locations page	Four sites instead of seven
Expansion detection	Regional engine	A municipal planning notice on a council site	Suggestive job postings and no decisive indicator
Claim provenance	Bing and archives	An earlier appearance and a removed citation	Fourteen apparently independent sources
Literature base	Baidu and Yandex	A fifth of the relevant literature, in two other languages	A monolingual conclusion presented as the field's position
Public records	Mojeek	An archived page defining a records system's coverage dates	A false negative on pre-migration records
International entity	Baidu, Yandex, national engine	The majority of material in three of four environments	Three subsidiaries instead of seven

Two patterns run through all seven. **In six of seven, the decisive finding came from something other than the largest index**, most often a small independent index, a regional engine, or an archive. And **in every case the value of the additional engines was coverage rather than ranking**: the material

existed and one crawler had it and another did not, which is Rule 10 demonstrated seven times.

Part 4: Chapter Summary

A search engine has four stages and each discards information. Crawling reaches a subset of the web, indexing keeps a subset of that, ranking orders it, and retrieval shows you ten results. What you see is four filters away from what exists, and each stage has a different failure mode with a different remedy: go to the source system, try another index, improve the query, or change your locale and personalization.

Two engines differ for five reasons: different indexes, different ranking, different localization, different personalization, and different freshness policies. A sixth matters for documentation: the same engine returns different results to the same person at different times, so **a search is not reproducible**, which is why the search log records engine, exact query, operators, locale, signed-in state, date and time, and result count, and why a results-page screenshot is itself evidence.

Rule 10 governs the chapter. Absence of results is a statement about the index, not about the world. An empty result is consistent with the page never being crawled, being crawled and not indexed, being indexed and ranked below where you looked, being withheld by your locale or personalization, sitting behind a query form or a login, having been removed, or simply not existing. Only one of those is "it does not exist," and a professional negative finding says which possibilities the search actually excluded.

Index independence determines whether agreement is corroboration. Engines with their own crawlers offer genuine second opinions; interfaces drawing on a partner index offer different presentation and the same coverage. Three engines agreeing while sharing one index is one source seen three times, which is Chapter 2's circular reporting in a new form. Because these arrangements change without notice, verify them yourself: check the engine's documentation, compare result sets on an obscure query, and record the date of your verification.

Query construction is most of search skill. Start narrow and widen, use the terms the source would use rather than the terms of your question, change one variable at a time, use exact phrases for distinctive strings, search for the artifact rather than the topic, and imagine the page that would answer your question before you search for it. Operators are concepts whose syntax varies by engine and changes without notice, they are sometimes accepted and silently ignored, and **no operator expands the index**.

The engines have distinct characters. Google offers the largest index, the best operators, unmatched document discovery, and specialized corpora, at the cost of heavy personalization, freshness bias, and restricted result depth. Bing is the principal independence check, is more generous to older and less optimized sites, and propagates into interfaces that appear independent. DuckDuckGo offers no personalization and therefore reproducible results, with coverage that may not be independent. Brave offers a genuinely independent index and, where provided, an indicator of how much of a result set is its own. Yandex offers the strongest visual similarity matching available and deep regional coverage,

with access and policy considerations. Baidu holds the deepest index of Chinese-language content and requires Chinese-language querying to be useful. Mojeek is small, fully independent, and reaches long-tail sites the majors ignore.

Regional engines matter because global engines are optimized for global queries. National engines index domestic sites, local-language content, and local conventions far more deeply, and in several markets the important asset is a national portal ecosystem rather than a search engine at all. Working with an engine you cannot read requires careful term selection from target-language documents, several variants, the native interface, machine translation for triage only, verification by a speaker for anything relied upon, and attention to transliteration, which multiplies the query set.

The multi-engine method exists for coverage first and corroboration second. Work through a major index, a second major index, an independent smaller index, a regional engine where the subject has a region, specialized corpora, archives, and then structured sources, recording what each uniquely contributed. Stop when a further engine adds nothing, and then change the query rather than the engine. Across all seven case studies, **the decisive finding came from something other than the largest index in six of them**, and in every case the value added was coverage rather than ranking.

Search engines find systems; systems hold records. Many questions put to a search engine are answered definitively by a registry, a docket, or a permit database, and the engine will give a worse answer more slowly. The competent move is frequently to use the engine to find out which system holds the answer, and then to leave the engine behind.

And the output of multi-engine searching is a coverage statement, naming the engines, the term sets, the locales, the dates, what each contributed uniquely, and what none of them returned, with the explicit note that a null result does not establish absence. That statement belongs in the product far more often than it appears.

Part 5: Key Terms

Grouped by theme. Terms carried from earlier chapters are marked and restated briefly.

5.1 Mechanics

Crawler (spider, bot). A program that fetches web pages by following links, subject to its own priorities and to site directives.

Crawl budget. The finite attention a crawler allocates to a site, which determines how much of it is fetched and how often.

Robots directives. Instructions a site publishes asking crawlers not to fetch or index particular content; honoured by mainstream crawlers as a convention.

Index. The searchable store of processed page content built from crawled pages.

Indexing. The stage at which crawled content is parsed and stored, during which content may be dropped as duplicative, thin, or low value.

Ranking. Scoring and ordering indexed pages for a given query.

Ranking signal. Any input to the scoring: relevance, authority, freshness, location, language, device, engagement, and others.

Retrieval and presentation. Returning a page of results with snippets, features, and answer boxes.

Result depth. How far into a result set a user can page, increasingly restricted on major engines.

Reported result count. An engine's estimate of matching pages; an approximation, not a measurement.

Freshness bias. Ranking preference for recent content, which buries original historical material.

Personalization. Adjustment of results based on a user's history, account, or behaviour.

Localization. Adjustment of results based on apparent location.

Language detection. Inference of a user's preferred language, which weights results toward it.

Snippet. The extract shown beneath a result, generated by the engine and not necessarily the page's own summary.

Cache. An engine's stored copy of a page; increasingly withdrawn as a feature.

5.2 Engine Types

Search engine. A service that crawls, indexes, ranks, and returns results.

Independent index. An index built by the operator's own crawler.

Licensed or partnered results. Results supplied substantially by another operator's index.

Hybrid index. Own crawl supplemented by a partner's results.

Metasearch engine. A service that queries several engines and merges the results, holding no index of its own.

Privacy proxy. A service that relays a query to another engine without passing the user's identifiers.

Portal ecosystem. A national platform whose blogs, forums, and knowledge services hold large volumes of content that general crawlers index poorly.

Specialized corpus. A purpose-built index of a content type, such as scholarly literature, patents, books, or news.

Vertical search. Search restricted to one content type or domain.

5.3 Method

Query construction. The practice of formulating searches, which is most of search skill.

Exact phrase search. Requiring words in a specified order; the highest-yield technique for names, titles, and distinctive strings.

Exclusion. Removing results containing a term.

Site restriction. Limiting results to a domain, subdomain, or top-level domain.

File type restriction. Limiting results to a document format; the primary document discovery technique.

Title, URL, and body restriction. Requiring a term in a specific part of a page.

Date restriction. Limiting results by time; the counter to freshness bias.

Wildcard. A placeholder for an unknown term within a phrase.

Operator deprecation. The silent weakening or removal of an operator, which requires periodic verification.

Silent operator failure. An engine accepting an operator and ignoring it, producing results that appear filtered and are not.

Artifact-oriented searching. Searching for the document that would contain the answer rather than for the topic.

Provenance tracing. Using exact-phrase search across engines to find the earliest appearance of a claim and detect copying.

Transliteration variants. The multiple romanizations of a name, each of which is a separate query.

Term selection from target-language documents. Taking query terms from a document in the language rather than from machine translation.

Coverage saturation. The point at which additional engines return nothing new, indicating that the query rather than the engine should change.

Coverage statement. The written record of which engines, queries, locales, and dates were used, what each contributed, and what none returned.

5.4 Analysis

Coverage boundary (Ch.3). The explicit statement of what a search did and did not cover, attached to every negative finding.

False negative (Ch.3). Concluding that something does not exist because a source with limited coverage did not return it.

Index independence. Whether two engines' results derive from separate crawls, which determines whether agreement is corroboration.

Circular reporting (Ch.1). Apparent corroboration created when sources trace to one origin; here, when engines share an index.

Unique finding. An item returned by exactly one engine, which is the measurable output of multi-engine searching.

Reproducibility (of a search). The degree to which a search can be repeated with the same result; low, and the reason for screenshots and logs.

Search log (Ch.6). The record of every query with its engine, terms, operators, locale, state, time, count, and outcome.

Regulatory filtering. Removal or suppression of results by legal requirement, which is a coverage boundary and an observation about the information environment.

Grey literature (Ch.3). Publicly available material that is not formally published or indexed, frequently reachable only by file type restriction.

Part 6: Knowledge Assessment

6.1 Short Answer (20 questions)

Answer in three to five sentences unless otherwise noted.

1. Name the four stages of a search engine's operation and state what each discards.
2. For each stage, give the reason a search might fail and the remedy.
3. Name five reasons two engines return different results for the same query.
4. Explain why a search is not reproducible, and what follows for documentation.
5. State Rule 10 and list five possibilities consistent with an empty result set.
6. Distinguish an independent index from licensed results, and explain why the distinction determines corroboration value.
7. Describe three ways to determine for yourself whether an engine has an independent index.
8. Distinguish a search engine, a metasearch engine, and a privacy proxy.
9. Explain personalization, localization, and freshness bias, and the mitigation for each.
10. Name six things a general crawler cannot reach, and connect them to Chapter 3.
11. State four principles of query construction and explain what "search for the artifact, not the topic" means.
12. Explain silent operator failure and how to detect it.
13. Explain why no operator can expand the index.
14. Describe Google's two greatest strengths and two greatest limitations for investigative work.
15. Explain why Bing matters structurally beyond its own user base.
16. Explain what DuckDuckGo provides that is valuable to investigators, and what it does not provide.

17. Explain why Brave Search is treated as a distinct entry rather than another privacy interface.
18. Explain why Yandex is used for reverse image work, and what considerations attach to using it.
19. Explain why searching Baidu in English produces little value, and what to do instead.
20. Explain why regional engines find material global engines miss, and describe the procedure for using an engine in a language you do not read.

6.2 Multiple Choice (15 questions)

1. Rule 10 states that: a) Every search must use at least three engines b) Absence of results is a statement about the index, not about the world c) Operators must be verified before use d) Automated querying is prohibited
2. A page exists on the web but appears in no search engine. The most likely explanation is that it: a) Does not exist b) Was never crawled, was crawled and not indexed, or is ranked below where you looked c) Has been deliberately hidden by the engines d) Requires an operator to find
3. Three engines return the same result. This constitutes corroboration only if: a) The result is highly ranked in all three b) The engines have independent indexes c) The query was identical d) The engines are from different countries
4. Which is the most reliable way to detect that two engines share an index? a) Comparing their privacy policies b) Comparing result sets for an obscure query and observing near-identical results and ordering c) Checking which country each is based in d) Counting the reported result totals
5. A search returns different results for two investigators running the same query. The least likely cause is: a) Different locales b) Different personalization c) Different search dates d) One investigator used a different browser window size
6. The primary counter to freshness bias when tracing a claim's origin is: a) Using more engines b) Date restriction combined with exact-phrase search c) Using a metasearch tool d) Increasing the result count
7. "Search for the artifact, not the topic" means: a) Search for images rather than text b) Search for terms that would appear on the document you want, such as "annual report" plus the entity name c) Search only file types d) Use the engine's specialized corpora
8. An operator is accepted by an engine but appears to have no effect on results. This is: a) Impossible b) Silent operator failure, and it should be tested on a query with a known answer c) Evidence the index is empty d) Normal and unimportant
9. Google's greatest single advantage for investigative work is usually: a) Privacy b) Index breadth combined with file type discovery c) Result depth d) Neutrality of ranking
10. Bing matters structurally because: a) It is the largest index b) Other interfaces have historically drawn on its index, so it propagates into apparently independent engines c) It does not personalize d) It has the best operators

11. DuckDuckGo's principal value to an investigator is: a) A larger index than Google b) No personalization, making results reproducible and comparable across investigators c) An independent crawler d) Specialized scholarly corpora
12. Brave Search is included as a distinct entry in this chapter primarily because it: a) Is the fastest engine b) Operates an independent index and, where provided, indicates how much of a result set is its own c) Has the strongest image search d) Covers regional languages best
13. Yandex is most valuable to investigators for: a) English-language corporate research b) Visually similar image matching, particularly for geolocation and altered images c) Scholarly literature d) File type discovery
14. Searching Baidu in English produces little value because: a) Baidu blocks foreign queries b) Its index is concentrated in Chinese-language content, so effective use requires Chinese-language queries c) English results are removed by regulation d) It has no operators
15. In the public records case study, the records themselves came from: a) Google b) Bing c) Querying the records portals directly, with engines used to find which systems exist d) An archive

6.3 True or False (10 questions)

Mark true or false. For each false statement, write one sentence explaining what makes it false.

1. If a search engine returns no results, the material does not exist on the web.
 2. A search performed today can be reproduced exactly by a colleague tomorrow.
 3. Two engines sharing an index provide independent corroboration.
 4. Operators can retrieve pages that a crawler never fetched.
 5. Personalization means two investigators may see different results for the same query on the same engine.
 6. A privacy-focused search engine necessarily has an independent index.
 7. Regional engines frequently index domestic and local-language content more deeply than global engines.
 8. Machine translation is adequate for triaging foreign-language results and inadequate for relying on them.
 9. The largest index will contain any page that any smaller engine has indexed.
 10. A reported result count is an approximation rather than a measurement.
-

Part 7: Critical Thinking Exercises

Selecting the appropriate engine

7.1 For each objective, name the two engines you would use first and the reason for each: (a) finding a company's investor presentations; (b) determining where a photograph of a building was taken; (c) researching a supplier registered in mainland China; (d) finding the earliest appearance of a circulating claim; (e) locating a small community organization's minutes from 2009; (f) checking whether a page you found on one engine is broadly indexed.

7.2 An investigator says they always start with the largest engine because it has the largest index. Assess this. When is it right, when is it wrong, and what would you propose instead?

7.3 You have thirty minutes and a question about a mid-sized manufacturer in a non-English-speaking European country. Write your engine sequence with time allocations and the reason for each step.

Comparing results across engines

7.4 Two engines return overlapping results for a query, with one returning three additional items. List four distinct explanations for the difference and describe how you would distinguish among them.

7.5 You run the same query on four engines. Three return near-identical result sets; the fourth returns a substantially different set. What is the most likely explanation for each group, and what does it mean for your coverage statement?

7.6 Write the coverage statement for a search across three engines in which one engine contributed two unique items and one query returned nothing anywhere. Use the form in Section 1.11.4.

Evaluating privacy considerations

7.7 You are researching a subject who, if they learned of the investigation, could plausibly obtain information about queries made about them. Assess which engines you would and would not use, and what else you would change.

7.8 A colleague argues that using a privacy-focused engine makes the investigation anonymous. Correct this, distinguishing privacy, anonymity, and coverage, with reference to Chapter 8.

7.9 Your organization prohibits the use of two of the engines in this chapter. Describe the coverage consequence, how you would state it in a product, and what you would do to mitigate it.

Identifying regional search opportunities

7.10 An entity operates in five countries. Describe how you would determine, for each, whether a regional engine or portal ecosystem exists and is worth using, before doing any substantive searching.

7.11 Your subject's name transliterates four ways. Describe the consequence for your search plan, how you would generate the variants, and how you would record the coverage.

7.12 A query returns substantive results on three engines and nothing on a fourth used in a particular market. Give three possible explanations, state how you would distinguish among them, and state how

you would report it.

Explaining differences between engines

7.13 Explain to a non-specialist client why your report cites results from four search engines rather than one, in two paragraphs, without technical jargon.

7.14 A client asks why a page they can see on their own site does not appear in search results. Give four possible reasons and the check for each.

7.15 Your colleague found an item on a small independent engine that appears in no major index. They are inclined to distrust it. Explain what its presence in one index and absence from others does and does not tell you about the item's reliability.

Part 8: Hands-On Lab Exercises

General Lab Guidance

Rules 1 through 10 apply. In particular: no private individuals as search subjects, no automated querying (Rule 5), Chapter 8's separate profile and signed-out discipline throughout, and every negative recorded with its coverage boundary (Rule 10).

Access and policy. Some engines may be unreachable from your network or restricted by your organization's policy. **Check before searching, not after.** Every lab has a path that works without any specific engine, and using a fallback path loses no marks provided you record why.

Search conditions. For all live searching: use a dedicated browser profile, signed out, with the locale and language settings recorded. Record every search to the Section 1.2.4 standard, including a screenshot of each results page you rely on.

Rubric scale. 0 to 4 per criterion: 4 professional, 3 competent, 2 developing, 1 attempted, 0 missing or fundamentally incorrect.

Lab 1: The Same Query Across Seven Engines

Time required: 2.5 to 3 hours **Format:** Individual **Difficulty:** Foundational to intermediate

Objectives

1. Execute identical queries across multiple engines under controlled conditions.
2. Document differences in ranking, coverage, freshness, language support, and relevance.
3. Identify unique results and assess index independence from observed behaviour.
4. Produce a documented comparison that another investigator could repeat.

Prerequisites

- Sections 1.1 through 1.9, and comparison tables 2.1 through 2.4.
- A dedicated browser profile (Chapter 8, Lab 1).

Choose a Track

Track	Requirement
A: Live	Access to the engines you may lawfully and properly use. Use as many as policy permits; record which you could not use and why
B: Provided observations	Use the observation set in Section 8.1.2. Loses no marks

8.1.1 The Query Set

Run all four queries on every engine available to you. **All four concern organizations, technical topics, or historical events; none concerns a private individual.**

Query	Type	What it tests
Q1: An exact phrase from a well-known public document, in quotation marks	Exact phrase	Index depth and provenance behaviour
Q2: The name of a small local institution in a non-English-speaking country, plus a generic term in that language	Regional and language	Language and regional coverage
Q3: A file type restriction combined with an organizational domain, seeking documents	Operator	Operator support and document discovery
Q4: A technical term that was current more than fifteen years ago and is now obsolete	Freshness and historical	Freshness bias and long-tail retention

Choose the specific queries yourself, record them exactly, and use the identical string on every engine. Do not adjust the query per engine; adjusting is a separate experiment and confounds this one.

8.1.2 Provided Observation Set (Track B)

Fictional but representative observations from one machine, one afternoon, one locale, signed out. "Overlap with Engine A" measures how many of the top ten results also appeared in Engine A's top ten.

Engine	Q1 top-10 results	Overlap with Engine A	Unique in top 10	Q2 relevant results	Q3 operator honoured?	Q4 top-10 age profile
A (largest index)	10	baseline	baseline	3 of 10	Yes, fully	7 items under 2 years old
B (second major)	10	4	5	2 of 10	Yes, fully	5 items under 2 years old
C (privacy interface)	10	3	1	2 of 10	Partially; results not fully filtered	5 items under 2 years old
D (independent, newer)	10	3	4	1 of 10	Partially	4 items under 2 years old
E (regional, other-language market)	10	1	8	9 of 10	Different syntax required	3 items under 2 years old
F (regional, other-script market)	10	0	10	0 of 10 (English query)	Different syntax required	2 items under 2 years old
G (small independent)	6	1	5	0 of 10	Partially	1 item under 2 years old; 4 items over 10 years old

Additional observations provided. Engine C's result set overlapped Engine B's by 8 of 10, substantially more than it overlapped Engine A. Engine G returned only six results for Q1 and reported no estimate of total matches. Engine E's Q2 results were almost entirely in the local language and included three domestic news sites that appeared in no other engine. Engine F returned nothing useful for an English-language Q2 and would require a query in its market's language.

Steps

Step 1 (20 min). Set conditions and record them on Worksheet 1A: profile, signed-out state, locale, language, date, time zone, and any engine you cannot use with the reason. Then write your four queries exactly as they will be entered.

Step 2 (60 min, Track A). Execute. For every engine and every query: record the top ten results by title and domain, the reported result count, and whether operators appeared to work. Screenshot each results page.

Step 3 (30 min). Overlap analysis on Worksheet 1B. For each engine pair, how many of the top ten results are shared? Build a matrix. Identify which pairs are strikingly similar and what that suggests about index sourcing.

Step 4 (25 min). Unique findings on Worksheet 1C. For every result that appeared on exactly one engine, record it, the engine, and an assessment of whether it looks valuable. State the total unique-result count per engine.

Step 5 (20 min). Dimension analysis on Worksheet 1D: for each engine, rate ranking quality, coverage, freshness, language support, and relevance for these queries, with a one-line justification for each rating. Ratings without justifications score 1.

Step 6 (20 min). Rule 10 analysis. Identify every query that returned nothing on at least one engine. For each, write the negative finding properly, with its coverage boundary, and state what it does and does not establish.

Step 7 (20 min). Write the findings summary, one page: which engine contributed most, which contributed something no other did, which two engines appear to share an index and on what evidence, and what you would change about your engine selection for a real investigation.

Worksheet 1A: Conditions

Element	Value
Profile and signed-out state	
Locale and language settings	
Date, time, time zone	
Engines used	
Engines unavailable, and why	
Q1 to Q4, exactly as entered	

Worksheet 1B: Overlap Matrix

	A	B	C	D	E	F	G
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Worksheet 1C: Unique Findings

Result	Engine	Query	Valuable?	Why
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Worksheet 1D: Dimension Ratings

Engine	Ranking	Coverage	Freshness	Language	Relevance	Justifications
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Expected Outcomes

- Conditions fully recorded, including any engine that could not be used and why. This is a coverage boundary and it is graded.

- **The overlap matrix should identify at least one pair with strikingly high overlap.** In the provided set, engines C and B overlap 8 of 10, substantially more than C overlaps A, which is evidence that C draws on B's index. Students who conclude that C is an independent second opinion have missed the lab's central finding.
- **Engine G's Q4 profile is the second key observation:** a small independent index returning predominantly old results, where the majors return predominantly recent ones. That is freshness bias and long-tail retention made visible in one row.
- **Engine E's Q2 result (9 of 10 relevant, against 1 to 3 for the majors) is the regional finding,** and it should drive a recommendation about non-English work.
- **Engine F returning nothing useful for an English query is not a failure of the engine.** It is a demonstration that the query, not the engine, was wrong, and students should say so.
- Operator support should be recorded as observed, including partial and silent failures.
- Negative findings written properly, with the engine, terms, locale, date, and the explicit statement that a null result does not establish absence.

Discussion Questions

1. Which two engines appear to share an index, and what is your evidence? How confident are you?
2. One engine returned mostly old results for Q4. Is that a weakness or a strength, and for what kind of investigation?
3. Which engine produced the most unique results, and were they valuable? Are those the same question?
4. Q2 performed very differently across engines. What does that imply for an investigation of a non-English-speaking subject?
5. If you had to drop three engines from a real investigation, which three and why?

Grading Rubric: Lab 1 (56 points)

Criterion	Weight	4 points looks like
Conditions recorded	x2	All fields, including unavailable engines with reasons
Identical queries used across engines	x2	Exact strings, unadjusted
Result capture completeness	x3	Top ten per engine per query, with counts and screenshots
Overlap matrix	x3	Complete, with the high-overlap pair identified
Index-sharing inference	x4	Correct inference from overlap

Criterion	Weight	4 points looks like
		evidence, with confidence stated
Unique findings identification	x3	Per engine, with a value assessment distinct from a count
Dimension ratings with justifications	x3	All five dimensions, justified
Freshness observation	x2	Q4 age profile differences identified and interpreted
Regional observation	x2	Q2 differences identified and interpreted
Rule 10 negatives	x3	Properly written with coverage boundaries
Findings summary	x2	Actionable engine selection recommendation

Total possible: 116 weighted. Passing: 80.

Lab 2: Researching a Company Across Four Engines

Time required: 2.5 to 3 hours **Format:** Individual **Difficulty:** Intermediate

Objectives

1. Compare what different engines contribute to the same investigative question.
2. Identify unique findings and attribute them to engines.
3. Assess which engine provided the most investigative value, distinguishing volume from value.
4. Produce a coverage statement.

Prerequisites

- Sections 1.1 through 1.11, and comparison tables 2.4 and 2.10.
- Lab 1 complete.

The Subject

Halgren Kiln Works, a fictional mid-sized industrial ceramics manufacturer. Everything below is invented; the company does not exist, which is why this lab uses a provided result packet rather than live searching.

The question. A prospective customer wants to know: how many production sites does Halgren operate, what is its ownership structure, and is there any public indication of operational or regulatory

difficulty?

8.2.1 The Provided Result Packet

Simulated results as four engines returned them. Each row is one result. "Rank" is its position in that engine's top twenty.

Engine A (largest index)

Rank	Result	Type
1	Halgren Kiln Works official site, home page: "three production facilities"	Company site
2	Halgren Kiln Works, About page	Company site
3	Industry directory listing: Halgren Kiln Works, two sites listed	Directory
4	Trade press: "Halgren expands capacity at Redmarsh" (14 months ago)	Trade press
6	Halgren product catalogue PDF	Company document
9	Recruitment listing aggregator, two open roles	Job board
12	Supplier quality manual PDF, hosted on Halgren's domain, listing four sites	Company document
17	Local news: planning application at Redmarsh (2 years ago)	Local press

Engine B (second major index)

Rank	Result	Type
1	Halgren Kiln Works official site	Company site
2	Industry directory listing	Directory
4	Trade press: Redmarsh expansion	Trade press
5	Regional business journal: "Halgren parent Vaskern Holdings restructures" (3 years ago)	Regional press
8	Halgren product catalogue PDF	Company document
11	Environmental regulator	Regulator

Rank	Result	Type
	bulletin listing an enforcement notice at a Halgren site (4 years ago)	
14	Recruitment aggregator	Job board
19	Trade association member listing, showing Halgren and two affiliated names	Association

Engine C (privacy interface)

Rank	Result	Type
1	Halgren Kiln Works official site	Company site
2	Industry directory listing	Directory
3	Trade press: Redmarsh expansion	Trade press
5	Regional business journal: Vaskern Holdings restructure	Regional press
7	Halgren product catalogue PDF	Company document
10	Environmental regulator bulletin	Regulator
15	Recruitment aggregator	Job board

Engine D (small independent index)

Rank	Result	Type
1	Halgren Kiln Works official site	Company site
3	Archived local chamber of commerce page (approximately 9 years old), listing Halgren at a fourth address	Local organization
5	A 2011 conference programme PDF listing a Halgren technical presentation, with an author affiliation to "Halgren Kiln Works (Torval division)"	Conference document
8	Halgren product catalogue PDF	Company document
12	A defunct regional industry newsletter, archived, describing a fire at a Halgren	Newsletter

Rank	Result	Type
	facility (approximately 12 years ago)	

Additional provided context. Engine C's result set overlaps Engine B's by six of seven items and Engine A's by four. Engine D reported no total result estimate. Engine A's reported total was approximately 4,100; Engine B's approximately 2,600; Engine C's approximately 2,600.

Steps

Step 1 (20 min). Extract findings on Worksheet 2A. Every distinct factual claim available from the packet, with the engine or engines that returned it and the result type. Do not interpret yet.

Step 2 (25 min). Unique attribution on Worksheet 2B. Which findings came from exactly one engine? Which from two? Which from all four? Produce a count per engine and, separately, an assessment of value per engine.

Step 3 (25 min). Index independence assessment on Worksheet 2C. Using the overlap figures and the reported totals, assess which engines appear to share an index and which are independent. State your confidence and your evidence.

Step 4 (30 min). Answer the three questions on Worksheet 2D: site count, ownership, and indications of difficulty. For each: the finding, the supporting items, the engines, the confidence level with its reason, and the gaps.

Step 5 (20 min). Conflict analysis. The packet contains at least three inconsistencies about the number of sites. Identify them, state what each source is and how authoritative it is, and describe how you would resolve the conflict, including which structured sources (Chapter 3) you would go to.

Step 6 (25 min). Write the coverage statement using the Section 1.11.4 form, plus a short assessment of which engine provided the most valuable intelligence and why volume and value diverged.

Step 7 (optional extension, 30 min). Repeat the exercise live against a real organization of your choosing, using four engines. Record the same worksheets. Bonus credit.

Worksheet 2A: Findings Extraction

#	Finding	Engines returning it	Result type	Authority
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Worksheet 2B: Unique Attribution

Engine	Unique findings	Total findings	Value assessment
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Worksheet 2C: Independence Assessment

Engine pair	Overlap evidence	Independent?	Confidence
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Worksheet 2D: Question Answers

Question	Finding	Supporting items	Engines	Confidence and reason	Gaps
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Expected Outcomes

- **The site count conflict is the central analytic content.** The company site says three, the industry directory says two, the company's own supplier quality manual says four, a nine-year-old chamber page names a fourth address, and a conference programme names a "Torval division" that appears nowhere else. The correct answer is that the public record does not establish a site count, that the company's own documents disagree with each other, and that the resolution requires structured sources: property records, permits, and regulatory registrations per site.
- **The supplier quality manual (Engine A, rank 12) is the highest-value single item** and it sits below the first page. Students who only consider top-ten results will miss it, which is the ranking lesson.
- **Engine B contributed the ownership finding, the regulatory finding, and the affiliated names**, none of which Engine A returned in its top twenty. On value rather than volume, Engine B is the strongest contributor to the actual questions asked.
- **Engine C is not an independent contributor.** Overlapping B by six of seven, with an identical reported total, is strong evidence of a shared index. Students who count C's findings as corroboration have made the error the lab exists to teach.
- **Engine D contributed three unique historical items**, of which the twelve-year-old fire and the "Torval division" affiliation are genuinely valuable leads. Its total volume is lowest and its unique value is high, which is the volume-versus-value distinction.
- **Confidence should be low to moderate throughout**, because everything in the packet is web search output and none of it is a structured record.
- The coverage statement must include the explicit Rule 10 sentence.

Discussion Questions

1. Engine A returned the most results and Engine B answered more of the questions. Explain the difference between volume and value to a client.
2. Engine C overlaps Engine B by six of seven. What does that do to a coverage claim of "we searched four engines"?

3. The most valuable single item was ranked twelfth. What does that imply about how you search?
4. The site count cannot be resolved from this packet. Write the sentence you would put in a report.
5. Engine D's findings are all old. When is that exactly what you need?

Grading Rubric: Lab 2 (60 points)

Criterion	Weight	4 points looks like
Findings extraction completeness	x2	Every distinct claim, attributed, typed
Unique attribution	x3	Per engine, with counts and a separate value assessment
Independence assessment	x4	Engine C correctly identified as non-independent, with evidence
Site count conflict identified	x4	All conflicting sources named; conflict not resolved by preference
Deep-ranked item found	x3	The supplier quality manual identified as highest value
Question answers with confidence	x3	All three questions, with reasons and gaps
Structured source recommendation	x3	Names the records that would resolve the conflict
Volume versus value analysis	x3	Explicitly distinguished, with Engine B identified as most valuable
Coverage statement	x3	Section 1.11.4 form, including the Rule 10 sentence
Live extension (bonus)	+3	Same worksheets against a real organization

Total possible: 112 weighted, plus 3 bonus. Passing: 77.

Lab 3: Regional Research

Time required: 2.5 to 3 hours **Format:** Individual **Difficulty:** Intermediate to advanced

Objectives

1. Conduct research using a country-specific engine or portal ecosystem.
2. Compare local results with global engine results for the same question.

3. Apply the procedure for working in a language you do not read.
4. Document language, transliteration, and coverage boundaries.

Prerequisites

- Sections 1.7, 1.8, 1.10, and comparison tables 2.5, 2.8, and 2.9.
- Lab 1 complete.

Choose a Track

Track	Requirement
A: Live regional research	Access to a regional engine or portal you may lawfully and properly use. Check organizational policy first. Select a market whose engine is accessible to you
B: Provided comparison	Use the observation set in Section 8.3.2

8.3.1 The Question

Choose a **public institution or a company** in a non-English-speaking country. Not a private individual. Suitable subjects: a municipal transport operator, a regional utility, a public university, a port authority, a mid-sized manufacturer.

Your question: what does the public web say about this organization's recent operational changes, in its own language, that is not available in English?

8.3.2 Provided Comparison Set (Track B)

Fictional. The subject is a municipal transport operator in a non-English-speaking European country.

Source of results	Query language	Relevant results in top 20	Result types	Unique items
Global engine, English query	English	3	Wikipedia-style entry, an English tourism page, one wire story	0
Global engine, local-language query	Local	9	Operator's own site, two national news items, a transport ministry page, four local news items	2
Regional engine, local-language query	Local	17	The above, plus six municipal council documents, three	8

Source of results	Query language	Relevant results in top 20	Result types	Unique items
			local forum threads, a regional trade publication, and two archived operator pages	
Regional engine, English query	English	2	Two tourism pages	0
Portal ecosystem search (same market)	Local	11	Nine community forum threads discussing service changes, two blog posts by a transport enthusiast group	11

Additional provided observations. The municipal council documents included minutes recording a decision to reduce service on two routes, dated four months before any news coverage. The forum threads contained photographs of posted notices at stops, dated before the council decision was published. The operator's name appears in the local language, in an official English rendering used on its own site, and in two different transliterations used by international sources.

Steps

Step 1 (25 min). Preparation on Worksheet 3A: the subject, the country, the language, the engines and portals available and their policy status, and **the name variants**, including the local-language form, the official English form, and every transliteration you can identify. Minimum four variants.

Step 2 (20 min). Term selection on Worksheet 3B. For each of three concepts you need to search (for example: service change, budget, board decision), take the target-language term **from a document in that language**, not from machine translation alone. Record where you took it from. Then record the machine-translated version and note any difference.

Step 3 (45 min). Execute (Track A) or analyse (Track B) on Worksheet 3C: for each source of results, the query used, the language, the relevant result count, the types of result, and the unique items.

Step 4 (25 min). Comparison analysis on Worksheet 3D. What did the regional source return that the global engine did not, and what category of material was it? What did the global engine return that the regional did not? What did the portal return that neither did?

Step 5 (20 min). Verification and translation discipline on Worksheet 3E. For every item you would rely on: how you triaged it, whether you would need a speaker's verification before relying on it, and what you would record in the search log.

Step 6 (25 min). Write the findings and coverage statement, one page: what the regional research established that English-language research would not have, the coverage boundary including the name

variants used, and an explicit statement of what you cannot assess because of the language barrier.

Worksheet 3A: Preparation

Element	Value
Subject and country	
Language and script	
Engines and portals available; policy status	
Name variants (minimum four)	

Worksheet 3B: Term Selection

Concept	Target-language term	Taken from	Machine translation	Difference
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Worksheet 3C: Execution

Source	Query	Language	Relevant results	Types	Unique items
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Worksheet 3D: Comparison

Category of material	Global engine	Regional engine	Portal	Notes
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Worksheet 3E: Verification

Item	Triage method	Speaker verification needed?	Log entry
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Expected Outcomes

- **Four-plus name variants**, and a recognition that each is effectively a separate query. Students who search one form have introduced a false negative.
- **The language effect should dominate the engine effect.** In the provided set, the global engine with a local-language query returned three times what it returned in English, before any regional engine was involved. That is the most important single finding: **query language matters more than engine choice**, and students frequently expect the opposite.
- **The regional engine's unique contribution is a category, not just a count:** municipal council documents, local forum threads, and regional trade material. Students should name the category.

- **The portal contribution is entirely community content**, which no crawler indexes well, and which contained the earliest evidence, predating both the news coverage and the published council decision.
- **The chronology finding is the analytic payoff**: notices posted at stops, photographed by forum users, predate the published council decision, which is a Chapter 5 timeline observation available only from the portal.
- Verification discipline: machine translation for triage, speaker verification before reliance, and original-language terms recorded in the log.
- A coverage statement that names the language barrier as a limitation rather than concealing it.

Discussion Questions

1. Which mattered more in your results: changing the engine or changing the query language? Why is that the more important lesson?
2. The portal returned material no crawler indexes. What does that imply about coverage claims in markets with strong portal ecosystems?
3. Photographs of posted notices predate the published decision. What does that establish, and what does it not?
4. You cannot read the language. State honestly what you cannot assess, and how you would word that in a product.
5. Which name variant produced the most results, and what would have happened if you had used only the English form?

Grading Rubric: Lab 3 (60 points)

Criterion	Weight	4 points looks like
Subject selection compliance	x1	Institution or company, not a private individual
Name variants	x3	Four-plus, including local form and transliterations
Term selection from target-language documents	x3	Terms sourced from real documents, differences from machine translation noted
Execution and recording	x3	Every source, query, language, and count recorded
Language-effect finding	x4	Identifies that query language outweighed engine choice
Regional unique contribution categorized	x3	Named as categories of material, not just counts
Portal contribution recognized	x3	Community content identified as

Criterion	Weight	4 points looks like
		uncrawlable
Chronology observation	x3	Notices predating the published decision identified and correctly limited
Verification discipline	x3	Triage versus reliance distinguished; speaker verification specified
Coverage statement with language limitation	x3	Honest and specific
Policy check recorded	x1	Engine availability and policy status stated

Total possible: 120 weighted. Passing: 82.

Lab 4: A Cybersecurity Incident Across Multiple Engines

Time required: 2.5 to 3 hours **Format:** Individual **Difficulty:** Advanced

Objectives

1. Use multiple engines to locate technical documentation, news coverage, public advisories, and archived content.
2. Attribute each finding to the engine that produced it and assess the contribution.
3. Distinguish authoritative technical sources from derivative coverage.
4. Produce a documented multi-engine coverage statement for a time-pressured investigation.

Prerequisites

- All of Part 1, and comparison tables 2.4, 2.7, and 2.10.
- Labs 1 through 3 complete.

The Incident

Fictional in all respects.

Your organization uses **Calderstone Fleet Manager**, a fictional vehicle telematics platform, from a fictional vendor, Calderstone Systems. This morning your security team learns from a peer that a flaw in the platform is being discussed publicly and may be exploited. Nothing internal has been detected.

Your task, within four hours: establish what the flaw is, which versions are affected, whether a fix exists, whether exploitation is reported, and what the vendor has said.

8.4.1 The Provided Result Packet

Simulated results from five engines and one archive. Each entry shows what the source returned and its rank where relevant.

Engine A (largest index)

Rank	Result	Type	Note
1	Calderstone Systems, product page	Vendor site	No security content
2	Trade press: "Fleet telematics flaw raises questions" (yesterday)	News	Derivative; cites "researchers"
3	Aggregator article repeating item 2	News	Derivative of a derivative
4	Social platform thread discussing the flaw	Social	Unverified
7	Calderstone Systems, support portal login page	Vendor site	Behind authentication
11	Calderstone Systems security advisory PDF, hosted on the vendor domain	Vendor advisory	Names affected versions 4.2 through 4.7
16	Industry newsletter summary	News	Derivative

Engine B (second major index)

Rank	Result	Type	Note
1	Calderstone Systems, product page	Vendor site	
2	Trade press item	News	
5	National computer emergency response team advisory	Authority	Independent severity assessment; recommends specific mitigations
6	Calderstone security advisory PDF	Vendor advisory	
9	A downstream vendor bulletin: an integrator that embeds Calderstone in its own	Vendor advisory	Names a different affected version range for its own product

Rank	Result	Type	Note
	product		
13	Social platform thread	Social	

Engine C (privacy interface)

Rank	Result	Type	Note
1	Calderstone product page	Vendor site	
2	Trade press item	News	
4	National CERT advisory	Authority	
6	Calderstone advisory PDF	Vendor advisory	
10	Downstream vendor bulletin	Vendor advisory	

Engine D (small independent index)

Rank	Result	Type	Note
2	A mailing list archive from 20 months ago describing the same behaviour, on a low-traffic domain	Technical discussion	Predates the disclosure
4	Calderstone product page	Vendor site	
6	A university research group's page listing a conference paper on telematics protocol weaknesses, naming Calderstone in the abstract	Academic	3 years old
9	An old Calderstone knowledge base article, still live, describing a configuration that the advisory now says is unsafe	Vendor site	Contradicts the advisory's recommendation

Engine E (regional engine, the vendor's home market, local-language query)

Rank	Result	Type	Note
1	Calderstone Systems local-language site	Vendor site	
3	Local-language vendor release notes, more detailed than the English advisory, naming versions 4.1 through 4.7	Vendor document	Wider range than the English advisory
5	Local business press: vendor comment on the incident	Regional press	Quotes a named executive
8	Local technical forum thread with a reported workaround	Forum	Unverified

Archive (not an engine)

Result	Note
Prior capture of the Calderstone advisory page, from 36 hours ago	Earlier version listed versions 4.4 through 4.7 only; the range was widened after publication
Prior capture of the knowledge base article, from 2 years ago	Content unchanged since

Steps

Step 1 (20 min). Source classification on Worksheet 4A. Every item in the packet, classified as: authoritative technical, authority or regulator, vendor primary, derivative news, community or unverified, or academic. Rank the classes by evidentiary weight and justify the ranking.

Step 2 (25 min). Engine contribution on Worksheet 4B. For each engine and the archive: what it uniquely contributed, what it duplicated, and the class of material it was strongest for.

Step 3 (30 min). The affected version question on Worksheet 4C. Four sources give version information and they do not agree. Record each, its authority, its date, and its scope, then state your assessment of the affected range with a confidence level, and state what you would do about the discrepancy.

Step 4 (25 min). Index independence on Worksheet 4D. Assess which engines appear to share an index, using the packet's overlap patterns, and state the consequence for a claim that five engines were searched.

Step 5 (25 min). Contradiction and change analysis. The knowledge base article contradicts the

advisory, and the advisory itself changed. Record both, state what each means, and state what you would tell your security team and the vendor.

Step 6 (30 min). Write the four-hour product: half a page, addressed to your CISO, with a bottom line, the affected version position with its uncertainty, the fix and mitigation position, the exploitation position, the coverage statement, and the immediate recommendations. Then attach the search log.

Worksheet 4A: Source Classification

Item	Class	Evidentiary weight	Reason
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Worksheet 4B: Engine Contribution

Engine	Unique contributions	Duplicated	Strongest for
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Worksheet 4C: Version Analysis

Source	Version range	Authority	Date	Scope	Notes
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Worksheet 4D: Independence

Engine pair	Evidence	Independent?	Consequence
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Expected Outcomes

- **The version discrepancy is the central finding.** The English advisory says 4.2 to 4.7; the local-language release notes say 4.1 to 4.7; the archived earlier advisory said 4.4 to 4.7; the downstream vendor gives a different range for its own product. The correct assessment treats the widest credible range as the working assumption, flags the discrepancy explicitly, and asks the vendor to confirm. Students who pick one range and report it as fact have failed the central criterion.
- **The advisory changed after publication,** and only the archive shows it. That is a Chapter 3 point demonstrated in a live incident, and it also indicates that the vendor's understanding is still developing, which matters for how much to rely on it.
- **The knowledge base article contradicts the advisory** and is still live. This is both an internal finding (the organization may have followed that guidance) and a responsible disclosure matter for the vendor (Chapter 7), and students should identify both.
- **Engine D's mailing list item, twenty months old, is the highest-value discovery item,** because it indicates the behaviour was known and discussed long before disclosure, which bears on exposure duration.

- **Engine C is again non-independent**, overlapping B closely with nothing unique. A claim of "five engines" is really three or four indexes.
- **Engine B contributed the CERT advisory and the downstream bulletin**, both of which are higher-authority than anything Engine A returned above rank 11.
- **The class ranking should place** authority and vendor primary sources above derivative news, with community content lowest and academic material useful for context rather than for version facts.
- The four-hour product must state uncertainty rather than resolve it artificially, and must include the coverage statement.

Discussion Questions

1. Four sources give four version ranges. Write the sentence you would put in the product.
2. The advisory changed 36 hours ago and only the archive shows it. What else does that tell you?
3. The knowledge base article contradicts the advisory. What are your two separate obligations here?
4. Engine D's finding is twenty months old. How does it change your assessment of exposure?
5. You searched five engines. How many indexes did you actually search, and how do you say that honestly?

Grading Rubric: Lab 4 (68 points)

Criterion	Weight	4 points looks like
Source classification and weighting	x3	All items classed; ranking justified
Engine contribution analysis	x3	Unique and duplicated separated per engine
Version discrepancy identified	x4	All four ranges recorded; not resolved by preference
Version position with uncertainty	x4	Widest credible range as working assumption, flagged, with vendor confirmation sought
Archive finding used	x3	Advisory change identified and interpreted
Knowledge base contradiction	x3	Both the internal and the disclosure obligation identified
Long-tail finding valued	x3	Mailing list item recognized as bearing on exposure duration
Independence assessment	x3	Engine C identified; the five-engines claim corrected

Criterion	Weight	4 points looks like
Four-hour product quality	x3	Bottom line, uncertainty, coverage statement, recommendations
Search log	x2	Complete to the Section 1.2.4 standard

Total possible: 124 weighted. Passing: 85.

Lab 5: Building a Search Engine Decision Matrix

Time required: 2 to 2.5 hours **Format:** Individual, then group comparison **Difficulty:** Intermediate

Objectives

1. Build a reusable decision matrix for engine selection.
2. Define selection criteria covering objective, language, geography, privacy, and expected source type.
3. Validate the matrix against the previous labs.
4. Produce something you will actually use.

Prerequisites

- All of Part 1, all comparison tables, and Labs 1 through 4.

Steps

Step 1 (25 min). Criteria definition on Worksheet 5A. Define your selection criteria, minimum six, each with the values it can take. Suggested: investigative objective, subject language, subject geography, expected source type, privacy sensitivity, time available, and independence requirement. For each, state why it belongs.

Step 2 (40 min). Build the matrix on Worksheet 5B. Rows are investigative situations; columns are engines. Each cell records: use first, use second, use if available, or do not use, with a compressed reason. Minimum twelve situations and minimum six engines or engine types. Situations should be specific, for example "document discovery on an English-language corporate subject" rather than "corporate research."

Step 3 (20 min). Decision flow on Worksheet 5C. A short flowchart, in text, that takes an investigator from a question to a starting engine in no more than five decision points. It must include the step that asks whether a search engine is the right tool at all.

Step 4 (25 min). Validation. Take the four labs you have completed and run each through your matrix. Does it recommend what actually worked? Record every case where the matrix would have led you

wrong, and adjust it. **Record the adjustments; the validation record is graded more heavily than the original matrix.**

Step 5 (20 min). Constraints layer on Worksheet 5D: how organizational policy, engine availability, language capability, and time available modify the recommendation. Include what an investigator does when the matrix recommends an engine they cannot use.

Step 6 (20 min). Write the usage note, half a page: how to use the matrix, its limitations, when to depart from it, and **its expiry**, meaning what would make it out of date and how often it should be reviewed.

Worksheet 5A: Criteria

Criterion	Values	Why it belongs
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Worksheet 5B: Decision Matrix

Situation	Engine 1	Engine 2	Engine 3	Engine 4	Engine 5	Engine 6
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Worksheet 5C: Decision Flow

Text flowchart, five decision points maximum.

Worksheet 5D: Constraints Layer

Constraint	Effect on the recommendation	Fallback
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Expected Outcomes

- Six-plus criteria, each justified. Students who omit language or independence have missed two of the chapter's main findings.
- Twelve-plus specific situations. Generic rows produce a matrix nobody uses.
- **The decision flow must include the "is a search engine the right tool" step**, and it should appear early. A matrix that routes every question to an engine has embedded the chapter's most common error.
- **The validation record is the point of the lab.** Every student's first matrix gets something wrong when tested against Lab 2 or Lab 3, most commonly by underweighting query language or by treating a non-independent engine as a second opinion. Recording the correction is worth more than getting it right first time.
- A constraints layer that handles the unavailable-engine case with a genuine fallback and a coverage statement, rather than ignoring it.
- An expiry statement, because the landscape changes and a matrix without a review date will be

used long after it is wrong.

Discussion Questions

1. Which criterion did you nearly omit, and what would that have cost?
2. Where did validation prove your matrix wrong?
3. Compare your matrix with a colleague's. Where do you disagree, and is the disagreement about facts or about weighting?
4. Your matrix will be out of date within a year. What specifically will change first?
5. What does your matrix say when the recommended engine is prohibited by policy?

Grading Rubric: Lab 5 (52 points)

Criterion	Weight	4 points looks like
Criteria definition	x3	Six-plus, justified, including language and independence
Matrix specificity	x3	Twelve-plus specific situations, six-plus engines
Cell reasoning	x3	Compressed reasons, not bare rankings
Decision flow	x3	Five points maximum, including the is-it-the-right-tool step
Validation against the labs	x4	All four labs run through; errors found and recorded
Adjustments recorded	x3	Changes made and the reasoning kept
Constraints layer	x2	Policy, availability, language, and time, with fallbacks
Usage note and expiry	x2	Limitations, departure guidance, and a review cadence

Total possible: 92 weighted. Passing: 63.

Part 9: Capstone Assignment

The Marrow Bay Terminal Investigation

Weight: 35% of chapter grade **Time required:** 6 to 9 hours **Format:** Individual **Deliverables:** Six items, specified in Section 9.3

9.1 The Scenario

Everything below is fictional.

You are engaged by **Corran Maritime Insurance**, a fictional specialist insurer, through its underwriting director. Corran is considering a substantial facility for **Marrow Bay Terminal Holdings**, a fictional bulk cargo terminal operator, and must decide within four weeks.

The subject. Marrow Bay Terminal Holdings is registered in an English-speaking jurisdiction. It states that it operates two terminals, one in its home market and one in a non-English-speaking European country. Its ultimate parent is stated to be a holding company in a third jurisdiction. It has a technical services agreement with a supplier in mainland China and, per its own website, a maintenance partnership with a company in a Russian-speaking market.

The underwriting director's questions.

1. Does the public record support the stated corporate structure and terminal count?
2. What does the public record show about the operational and regulatory history of each terminal?
3. Are there indications of incidents, disputes, or enforcement at any facility?
4. What can be established about the supplier and partner relationships?
5. What cannot be established from open sources, and what would resolve it?

Constraints.

- Four weeks; you are the only investigator.
 - Your engagement covers open source research only, with no contact with any party.
 - Your firm's policy requires approval before using two of the engines in this chapter; assume approval is available on request and that the request must be documented.
 - You read English and have basic reading ability in one other European language of your choosing.
 - Rules 1 through 10 apply. In particular, no private individuals: officers appear only in their official capacity from official records.
-

9.2 Guidance on the Substance

Four information environments, four languages, and at least four indexes. An investigation conducted entirely in English on one engine would answer perhaps a third of the questions and would not know which third.

Question 5 is the most heavily weighted. An honest account of what open sources cannot establish, with the structured sources or non-open routes that would resolve each gap, is worth more to an underwriter than a confident answer built on thin coverage.

Search engines are not the source. Chapter 3 applies throughout: the engines find the systems, and the registries, dockets, permit databases, and regulator portals hold the answers. A capstone that treats search results as findings rather than as leads to structured sources will score poorly.

9.3 What You Must Produce

Deliverable 1: Search Strategy and Engine Justification (20 points)

- The decomposition of the five questions into searchable propositions, minimum twelve.
- For each information environment (home market, European market, mainland China, Russian-speaking market): the engines and portals selected, with a stated reason per selection, and the engines rejected with reasons.
- **An index independence analysis:** which of your selected engines are genuinely independent, how you determined it, and what your coverage claim will therefore be.
- The name and transliteration variant list for every entity, minimum twelve variants across all entities.
- Term selection for each non-English environment, with terms taken from target-language documents where possible.
- The policy approval record for any restricted engine, or the recorded decision not to seek it and the coverage consequence.

Deliverable 2: Search Execution Log (16 points)

Minimum forty logged searches to the Section 1.2.4 standard. Every entry: search ID, engine and interface, exact query, operators and whether they worked, locale and language settings, signed-in state, date and time with time zone, reported result count, findings by item ID, and negatives with coverage boundaries.

You may execute these searches live against **real, public, institutional sources of your choosing that stand in for the fictional entities** (for example, searching a real port operator to demonstrate the method), or you may document the searches you would run with the expected result types. **State which you did.** Either is acceptable; the method and documentation are what is graded.

Deliverable 3: Findings by Question, with Source Attribution (20 points)

For each of the five questions:

- The findings, each with the item, the engine or source that produced it, its class (authoritative, vendor or subject primary, regulator, press, community, archive), and its reliability and credibility ratings (Chapter 1).
- Which findings are unique to one engine.
- Which findings came from structured sources reached via a search engine rather than from the

engine itself.

- Conflicts between sources, recorded and not resolved by preference.
- A confidence level per question, with the reason.

Deliverable 4: Multi-Engine Coverage Statement (14 points)

- Engines and portals used, per environment, with dates and locales.
- What each contributed uniquely.
- Which engines shared an index, and the corrected count of indexes actually searched.
- Every query that returned nothing anywhere, written as a proper negative with its boundary.
- **An explicit Rule 10 statement.**
- The language coverage limitation, stated honestly, including what you could not assess.

Deliverable 5: Investigative Report (24 points)

Maximum four pages plus annexes. Addressed to the underwriting director.

Required: a bottom line of no more than four sentences with a judgment and a confidence level; answers to the five questions, each with facts separated from assessments and every fact cited to an item; **the answer to question 5 treated as a substantive finding rather than a caveat**, listing each gap, why open sources cannot close it, and precisely what would (a structured source, a records request, a non-open route, or a direct enquiry the engagement does not permit); the source reliability assessment; assumptions; ranked gaps; at least six recommendations that are specific and assignable; and what would change the assessment. Annexes: the search log, the coverage statement, and a source inventory.

Deliverable 6: Methodology Reflection (6 points)

Two pages maximum: which engine contributed most and whether that matched your expectation; where query language outweighed engine choice; where the multi-engine approach was wasted effort; what you would do differently; and one specific change you will make to your own search practice.

9.4 Capstone Grading Rubric (110 points, scored to 100)

Deliverable	Criterion	Points
1. Strategy	Twelve-plus searchable propositions	4
	Engine selection per environment with reasons	5
	Rejected engines with reasons	2
	Index independence analysis	5

Deliverable	Criterion	Points
	Twelve-plus name and transliteration variants	3
	Target-language term selection	3
	Policy approval recorded, or the consequence stated	2
2. Log	Forty-plus entries to the full standard	8
	Negatives with coverage boundaries	5
	Operator behaviour recorded, including silent failures	3
3. Findings	Findings attributed to engine and source class	5
	Reliability and credibility ratings	4
	Unique findings identified	3
	Structured-source findings distinguished from engine findings	4
	Conflicts recorded and not resolved by preference	2
	Confidence per question with reasons	2
4. Coverage	Per-environment engine and locale record	3
	Unique contribution per engine	3
	Shared-index correction to the engine count	4
	Proper negatives and the Rule 10 statement	2
	Language limitation stated honestly	2
5. Report	Bottom line: four sentences, judgment, confidence	3
	Five questions answered, facts separated and cited	5
	Question 5 treated as a substantive finding with resolving routes	6

Deliverable	Criterion	Points
	Source reliability assessment	3
	Assumptions and ranked gaps	3
	Recommendations, specific and assignable	2
	Length discipline and annexes complete	2
6. Reflection	Honest, specific, with a practice change	6
Total		110, scored to 100

Grade bands: 90 to 100 exemplary, 80 to 89 proficient, 70 to 79 competent, 60 to 69 developing, below 60 not yet passing.

Automatic deductions:

- Claiming a coverage boundary broader than the engines searched support, or stating that something does not exist on the basis of null search results: **minus 15**. This is Rule 10 and it is the chapter's central test.
- Counting engines that share an index as independent sources in the coverage statement: minus 10.
- Conducting the investigation in English only, or failing to use target-language queries in the non-English environments: minus 12.
- Treating search results as findings without going to the structured sources they point to: minus 10.
- Any automated querying of a search engine: minus 20. Rule 5.
- Researching any private individual, including officers beyond their official capacity: minus 20.
- No index independence analysis: minus 8.
- Missing search log, or a log below twenty entries: capped at 70.

Guidance on the expected shape of a strong package. The strategy should assign a different engine set to each of the four environments and should say why, with the home market covered by two major independent indexes plus a small independent one, the European market by a global engine with local-language queries plus the national engine, mainland China by a domestic engine with Chinese-language queries, and the Russian-speaking market by the regional engine with Russian-language queries. The independence analysis should reduce an apparent count of six or seven engines to four or five actual indexes. The transliteration list should be genuinely long, because a bulk terminal operator with operations in four language environments will appear under many forms, and a missed variant produces a false negative that looks like a finding. Question 5 should identify at least: beneficial ownership

behind the third-jurisdiction parent, which open sources may not reach; the financial condition of the supplier, which is unlikely to be public; incident history at the European terminal, which may exist only in a national regulator's system reachable by request rather than by search; and the substance of the maintenance partnership, which is likely contractual and private. And throughout, the report should show that the search engines found the registries, the dockets, the permit systems, and the regulator portals, and that those systems produced the findings.

Part 10: Answer Keys

Instructors may wish to remove this section from student copies. Engine behaviour changes; where an answer describes a partnership or a feature, treat it as a snapshot and verify.

10.1 Short Answer Key

1. Crawling fetches a subset of the web, discarding what is not discoverable, not permitted, or not prioritized. Indexing keeps a subset of what was crawled, discarding duplicates and material judged low value. Ranking orders the indexed pages, discarding by burying. Retrieval shows a page of results, discarding everything below it.
2. Crawling: go to the source system directly, or try an engine with a different crawler. Indexing: try another engine or an archive. Ranking: improve the query, use operators, or use an engine with different ranking. Retrieval: change locale and language settings, search signed out, or use a different engine.
3. Different indexes, different ranking, different localization, different personalization and session state, and different freshness policies.
4. Because indexes and ranking models change continuously, so the same query returns different results at different times, and personalization and location vary by user. It follows that the search log must record engine, exact query, operators, locale, signed-in state, date and time, and result count, and that a results-page screenshot is itself evidence.
5. Rule 10: absence of results is a statement about the index, not about the world. Consistent with a null result: never crawled, crawled but not indexed, indexed but ranked below where you looked, withheld by locale or personalization, behind a query form or login, removed since publication, in a deprioritized language or format, or genuinely non-existent.
6. An independent index is built by the operator's own crawler; licensed results are supplied by another operator's index. It matters because agreement across independent indexes means two crawlers found the material, which is weak but real corroboration of publication, whereas agreement across engines sharing an index is one source presented several times.
7. Check the engine's own documentation for a described crawler; compare result sets on an obscure query and look for near-identical results, snippets, and ordering; and test for pages known to be in one index and not another.

- 8.** A search engine crawls and holds its own index. A metasearch engine queries several engines and merges results, holding no index. A privacy proxy relays your query to another engine without your identifiers, providing that engine's coverage with better privacy.
- 9.** Personalization adjusts results by your history and account; mitigate by searching signed out in a clean profile. Localization adjusts by apparent location; mitigate by setting region explicitly and recording it. Freshness bias boosts recent content and buries original historical material; mitigate with date restriction and archives.
- 10.** Content behind query forms, behind authentication, excluded by directives, unrenderable or unparseable by the crawler, unlinked, removed since publication, and in deprioritized languages or formats. These are Chapter 3's deep web, restated mechanically.
- 11.** Any four of: start narrow and widen; use the terms the source would use; change one variable at a time; use exact phrases for distinctive strings; search for the artifact rather than the topic; anticipate the page that would answer the question; iterate and record. "Search for the artifact" means searching for the words that would appear on the document you want, such as a report title plus an entity name, rather than describing your question.
- 12.** Silent operator failure is an engine accepting an operator and ignoring it, producing results that appear filtered but are not. Detect it by testing the operator on a query whose correct answer you already know.
- 13.** Because operators filter and reorder what is already in the index. If the crawler never fetched a page, no query formulation can return it.
- 14.** Strengths: the largest index and unmatched document discovery via file type restriction, plus specialized corpora. Limitations: heavy personalization whose effect you cannot see, freshness bias that buries historical material, restricted result depth, and the most extensive data collection.
- 15.** Because other interfaces have historically drawn on its index, so its coverage propagates into engines that appear independent. An investigator checking Bing and two Bing-derived interfaces has checked one index three times.
- 16.** It provides no personalization, which makes results reproducible and comparable between investigators and across time, plus reduced referrer leakage and privacy for sensitive queries. It does not necessarily provide independent coverage, and it is not anonymity.
- 17.** Because it operates an independent index rather than relaying a partner's, which makes it a genuine third opinion on coverage, and because where it provides an indicator of how much of a result set came from its own index, that indicator is directly useful to an investigator assessing independence.
- 18.** For visually similar matching, which is materially stronger than the majors and is what geolocation work requires, since cropped, altered, and re-encoded images frequently fail identical-match search. Considerations: weaker coverage outside its core regions, access variability, organizational policy, and processing under a different legal regime.
- 19.** Because its index is concentrated in Chinese-language content, so English queries return a small and unrepresentative slice. Instead, query in Chinese using terms taken from Chinese-language

documents, try several term variants, and have anything material checked by a reader of the language.

20. Regional engines index domestic sites, national-language content, and local conventions more deeply, and in some markets reach inside portal ecosystems that general crawlers cannot. The procedure: take terms from target-language documents, try several variants, use the native interface, translate results for triage only, have anything relied upon verified by a speaker, record original-language terms in the log, and treat each transliteration as a separate query.

10.2 Multiple Choice Key

#	Answer	Note
1	b	Rule 10
2	b	Any of the three index-side explanations
3	b	Independent indexes
4	b	Near-identical result sets on an obscure query
5	d	Window size is not a meaningful ranking factor here
6	b	Date restriction plus exact phrase
7	b	Search the words on the document you want
8	b	Silent operator failure; test on a known answer
9	b	Index breadth plus file type discovery
10	b	Downstream propagation of its index
11	b	No personalization, so results are reproducible
12	b	Independent index with an independence indicator
13	b	Visually similar matching
14	b	Index concentrated in Chinese-language content
15	c	The portals, found via the engines

10.3 True or False Key

#	Answer	If false, why
1	False	A null result is consistent with never crawled, not indexed, buried by ranking, withheld by locale, behind a form or login, or removed
2	False	Indexes and ranking change continuously, and results vary by locale and personalization
3	False	Shared-index agreement is one source seen twice
4	False	Operators filter the index; they cannot add to it
5	True	
6	False	Several privacy-focused engines draw substantially on a partner's index
7	True	
8	True	
9	False	Smaller independent crawlers routinely index long-tail pages the majors never fetched
10	True	

10.4 Critical Thinking Guidance

7.1 (a) A major index with file type restriction, plus a second major index for depth. (b) The engine with the strongest visually similar matching, plus a second general engine for identical matches. (c) A domestic Chinese engine with Chinese-language queries, plus a global engine for any international footprint. (d) A major index with date restriction plus a small independent index with date sorting, and archives. (e) A small independent index for long-tail retention, plus archives. (f) A second independent index, because the question is precisely one of index coverage.

7.2 Right when index breadth is the binding constraint and the subject is English-language and well covered. Wrong when the subject is non-English, regional, historical, or long-tail, and wrong whenever the question is about coverage rather than about finding one page. The proposal: start with the environment and language, then choose the engine.

7.3 A defensible allocation: five minutes on name and term variants, five on the global engine in English for orientation, ten on the global engine in the local language, ten on the national engine in the local language, and the balance on structured sources the searches identified. The reasoning should

show that language change comes before engine change.

7.4 Four explanations: different indexes; different ranking with the items present in both but buried in one; different localization; and different freshness, with one engine having recrawled more recently. Distinguish by searching for the specific items directly on the engine that omitted them, which separates ranking from indexing.

7.5 The three similar sets most likely share an index or draw on a common source; the outlier most likely has an independent index. For the coverage statement, the four engines represent two indexes, and the claim must say so.

7.6 Should follow Section 1.11.4's form and must include the Rule 10 sentence and the specific null query with its terms, locale, and date.

7.7 Prefer engines that do not profile searches and do not retain query history against an identity; avoid engines whose data the subject could plausibly obtain. Also change: search from a separate profile signed out, use archives rather than live visits to the subject's properties, and treat the query itself as a disclosure (Chapter 8).

7.8 Privacy is not profiling your queries; anonymity is your identity not being connected to the activity; coverage is what the index holds. A privacy-focused engine improves the first, does nothing for the second, since your IP address, fingerprint, and network are unchanged, and may do nothing for the third if it draws on a partner index.

7.9 The consequence is reduced coverage in specific environments, most likely regional and non-English material and image similarity work. State it in the coverage statement as a named limitation with the environments affected. Mitigate by using the permitted engines in the target language, by using national portals and structured sources directly, and by seeking approval where the case justifies it.

7.10 For each country: check whether a national engine with an independent index exists and is accessible; check whether a portal ecosystem holds significant content; check what proportion of relevant content is likely in the national language; and check organizational policy. Then decide per country rather than applying one approach to all five.

7.11 Each variant is a separate query, and the coverage claim must name the variants used. Generate them from official records in the local language, from the entity's own materials, from international press renderings, and from any standard transliteration schemes. Record all of them, including those that returned nothing.

7.12 Three explanations: the engine's index does not cover the material; the query terms are wrong for that language or market; or the results are filtered by regulation. Distinguish by testing a query whose answer you know in that market, by trying alternative term variants, and by checking whether adjacent topics return results. Report it as a coverage observation, and where filtering is the likely cause, report that as an observation about the information environment rather than about the subject.

7.13 Should explain, without jargon, that different search engines have looked at different parts of the web, that using one means accepting one company's view of what exists, and that the report names

what each contributed so the reader can see the basis.

7.14 Four reasons: the page is not linked from anywhere the crawler reaches; the site asks crawlers not to index it; it was crawled but judged duplicative or low value; or it is indexed and ranked below where they looked. Checks: search for a distinctive exact phrase from the page; check the site's crawler directives; check whether a near-identical page exists elsewhere on the site; and search with a site restriction.

7.15 Its presence in one index and absence from others tells you about crawler coverage and priorities and nothing whatever about the item's reliability. A small engine indexing an obscure but genuine page is exactly what small engines are for. Reliability is assessed the same way as for any source: the production process, the publisher, the date, and corroboration (Chapters 1 and 3).

Appendix A: Reusable Templates

A.1 Search Log Entry

Field	Value
Search ID	
Engine and interface (including country version)	
Exact query string	
Operators used	
Operators appeared to work?	
Locale setting	
Language setting	
Date range filter	
Signed-in state and profile	
Date, time, time zone	
Reported result count	
Items found (by item ID)	
Nothing found? Coverage boundary	
Results page screenshot reference	
Searcher	

A.2 Multi-Engine Coverage Statement

ENGINES AND INTERFACES USED: _____
DATES: _____ LOCALE(S): _____ LANGUAGE(S): _____
SIGNED-IN STATE: _____

TERM SETS: (reference the search log) _____

NAME AND TRANSLITERATION VARIANTS USED: _____

UNIQUE CONTRIBUTIONS

Engine 1: _____

Engine 2: _____

Engine 3: _____

INDEX INDEPENDENCE

Engines searched: _____ Distinct indexes searched: _____

Basis for this assessment: _____

QUERIES RETURNING NOTHING ANYWHERE: _____

RULE 10 STATEMENT: No engine returned any result matching [query] under these terms, locales, and dates. This does not establish that no such material exists; it establishes that it is not indexed by these engines under these terms as of [date].

LANGUAGE LIMITATION: _____

POLICY LIMITATION (engines unavailable): _____

A.3 Engine Selection Worksheet

Question	Answer
Is a search engine the right tool, or does a structured system hold this?	
Subject language(s)	
Subject geography	
Expected source type (document, press, record, image, community)	
Independence requirement	
Privacy sensitivity of the query	
Time available	
Policy constraints	
Engines selected, in order, with reasons	
Engines rejected, with reasons	

A.4 Overlap and Independence Matrix

	Engine A	Engine B	Engine C	Engine D	Engine E
Engine A					
Engine B					
Engine C					

Cell value: shared results in the top ten. High overlap plus similar snippets and ordering suggests a shared index.

A.5 Name and Transliteration Variant List

Entity	Local-language form	Official English form	Transliteration 1	Transliteration 2	Former names	Abbreviations	Searched?	Results?
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A.6 Negative Finding Template (search-specific)

Engine: _____ Interface/country version: _____
Exact query: _____
Operators: _____ Locale: _____ Language: _____
Date range applied: _____ Signed-in state: _____
Date/time of search (TZ): _____
Result: no relevant results returned

COVERAGE BOUNDARY: This engine indexes [what it is known to cover]. It does not cover [query forms, authenticated content, unlinked pages, removed pages]. Other engines, other name variants, and other languages were [searched / not searched].

THEREFORE THIS DOES NOT ESTABLISH: _____
NEXT STEP TO CLOSE THE GAP: _____

A.7 Decision Flow

1. Would a structured system answer this definitively?
YES → Go there (Chapter 3). Use an engine only to find which system.
NO → continue
2. What language and geography is the subject?
Non-English → query in the target language first, then add the national engine
3. What kind of artifact would answer it?
Document → major index with file type restriction
Image → the strongest visual-similarity engine
Old page → small independent index plus archives
Record → back to step 1
4. Do I need independence, or coverage, or both?
Independence → add a second independent index
5. Is the query itself sensitive?
YES → a non-profiling engine, in a separate profile, signed out

Appendix B: The Search Discipline Checklist

Run before, during, and after any search-based collection.

Before

1. Would a structured source answer this better and faster?
2. What language and geography does the subject actually operate in?
3. Which engines have independent indexes, and have I verified that recently?
4. What are the name and transliteration variants?
5. What are my locale and language settings, and are they recorded?
6. Am I signed out, in a dedicated profile?
7. Does policy permit every engine I plan to use?

During 8. Am I logging every query exactly as entered? 9. Am I recording result counts and screenshotting results pages I rely on? 10. Have I tested whether operators are actually working? 11. Am I changing one variable at a time? 12. Am I looking past the first page where the question warrants it? 13. Am I recording negatives with boundaries as I go?

After 14. Which findings are unique to one engine? 15. Which engines shared an index, and what is my corrected index count? 16. Have I gone to the structured sources my searches identified? 17. Is my coverage statement written, including the Rule 10 sentence? 18. Have I stated the language limitation honestly? 19. Would a colleague repeating this get the same result, and if not, have I said so? 20. Does any finding rest on search results alone that should rest on a record?

Appendix C: Where to Go Next

Skills to build after this chapter, in priority order:

1. **Verify the index landscape yourself, this month.** Run the obscure-query comparison across five engines and record which pairs look identical. Repeat annually, because the answer changes.
2. **Learn one non-English search environment properly**, including its national engine, its portal ecosystem, and its term conventions. Depth in one is worth more than familiarity with six.
3. **Practise the artifact-oriented query.** For a week, before every search, write down the document you are hoping to find, then search for its words.
4. **Build the operator test habit.** For each engine you use, test every operator you rely on against a query with a known answer, and record what you found.
5. **Practise provenance tracing** with exact-phrase search across engines, on any circulating claim, until finding the earliest indexed appearance is routine.

6. **Learn the structured sources for your domain** so thoroughly that you stop reaching for a search engine first.
7. **Keep your decision matrix alive.** Review it on the cadence you set in Lab 5, and record what changed.

Habits to start now:

- Ask whether a structured system holds the answer before you type anything.
 - Change the language before you change the engine.
 - Record the query, the locale, and the time, every time.
 - Screenshot the results page, not just the destination.
 - Treat every null result as a statement about an index.
 - Say how many indexes you actually searched, not how many engines.
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End of Chapter 9